

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250278749

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Mitchell; Stephen John et al.

INTEGRATED WEATHER GRAPHICAL USER INTERFACE

Abstract

Data integration and distribution systems. A system includes a graphical user interface (GUI). Weather and market data are collected. A weather symbology including symbol elements linked to segments of the collected weather data and rules for generating weather symbology instructions are stored. The GUI is generated for display on a user device. A weather symbology instruction is determined based on at least one requested symbol element indicated in a weather data request and the rules. A weather forecast dataset is created from among the collected weather data based on the weather symbology instruction. A presentation package including the weather forecast dataset and the collected market data is generated such that the weather forecast dataset is integrated with the collected market data. The presentation package is presented on the GUI and updated concurrent with changes to at least one of the weather data, the market data and user input.

Inventors: Mitchell; Stephen John (Houston, TX), McSween; Andrew Castellani (Chicago, IL), Batten; Marc (San Jose, CA)

Applicant: Intercontinental Exchange Holdings, Inc. (Atlanta, GA)

Family ID: 76316945

Assignee: Intercontinental Exchange Holdings, Inc. (Atlanta, GA)

Appl. No.: 19/213080

Filed: May 20, 2025

Related U.S. Application Data

parent US continuation 18411867 20240112 parent-grant-document US 12340382 child US 19213080

parent US continuation 18203350 20230530 PENDING child US 18411867

parent US continuation 17943715 20220913 parent-grant-document US 11775990 child US 18203350

parent US continuation 17696323 20220316 parent-grant-document US 11521219 child US

17943715
parent US continuation 17490015 20210930 parent-grant-document US 11308507 child US
17696323
parent US continuation 17186122 20210226 parent-grant-document US 11170390 child US
17490015
us-provisional-application US 62982235 20200227

Publication Classification

Int. Cl.: **G06Q30/0201** (20230101); **G01W1/00** (20060101); **G06F3/0482** (20130101);
G06Q40/04 (20120101)

U.S. Cl.:

CPC **G06Q30/0201** (20130101); **G01W1/00** (20130101); **G06F3/0482** (20130101);
G06Q40/04 (20130101);

Background/Summary

TECHNICAL FIELD

[0001] The present disclosure generally relates to improving data structure distribution and, in particular to distribution systems, interactive graphical user interfaces (GUIs) and methods for the integration of disparate data structures for interaction, including real-time integration of weather and market data structures.

BACKGROUND

[0002] Problems exist in the field of digital distribution platforms. In general, a digital distribution platform may manage digital data content (e.g., digital goods, digital information, etc.) and distribute the content to various end-users. Conventional platforms may distribute digital data content from one or more data sources (e.g., data feeds, data files, user input and the like) that may be distributed across one or more networks, may include different data types, different data formats, different data communication requirements, different network security, different availability time periods and the like. Moreover, distribution platform may distribute data content in one or more distribution formats (e.g., in a data file, on a user interface, in a spreadsheet and the like), to particular end-users with varying amounts of data and/or personalized data and the like. Conventional platforms also exist that may provide the ability for user-interaction with the distributed data (e.g., data analysis tools, actions that may be performed with the data and the like) in addition to the presentation of distributed digital data content.

[0003] All of the above variables associated with data distribution make it technically difficult to manage data distribution and interaction for real-time distribution. Yet further, distribution of digital data content in real-time becomes increasingly difficult as the volume of digital data content to be distributed increases and/or as the digital data content changes more rapidly over time (e.g., with increasing volatility of the data content). For example, it may become increasingly technically difficult for a distribution platform to continually update an interactive user interface with the most up-to-date data content, when the data volume increases and/or the data content itself changes rapidly. In such instances, any transmission delays over one or more networks to obtain the data content coupled with any data handling delays by the distribution platform for handling the received data content (e.g., to convert a data format of received data content, to normalize any data content, to remove any data content not suitable for presentation, to generate data for distribution in

one or more distribution formats, create aggregated output data, generate any user interfaces and the like) may introduce significant errors in distributed data and the ability by the end-user to interact with the distributed content. For example, the distributed data content may not provide the most up-to-date information, leading to a situation in which a user may perform an action based on stale information (e.g., an auction for an object at a price that no longer exists, respond to older data content when newer content exists, etc.).

[0004] Another significant technical problem that still exists in data distribution platforms includes the integration of data content for distribution that includes disparate data types. For example, while it may be possible to simply display disparate data types on user interface side-by-side, it may be difficult to integrate disparate data types in an intelligent manner, e.g., where the integration of the disparate data types provides meaningful information about the combination of disparate data types. For example, a first data type may include data values that are based on one set of underlying index values whereas a second (different and independent) data type may be based on a completely different and unrelated set of underlying index values. Thus, it may be technically difficult to suitably align and integrate disparate data types. It may be still more difficult to integrate disparate data types with real-time, rapidly changing data.

[0005] Accordingly, there is a need for a system (including a novel interactive GUI), and method for integrating and distributing disparate data types in a fully-automated (or near fully-automated) manner. All of this, without significant increases to the computational burden, cost, system complexity, re-programming requirements and system maintenance.

SUMMARY

[0006] Aspects of the present disclosure relate to data integration and distribution systems, methods and non-transitory computer readable medium. A system includes one or more weather data source systems, one or more market data source systems, at least one user device associated with at least one user and a data distribution system. The data distribution system is in communication with the one or more weather data source systems, the one or more market data source systems and the at least one user device via at least one network. The data distribution system includes an interactive graphical user interface (GUI) for interactively presenting integrated weather and market data. The data distribution system is configured to collect weather data among the one or more weather data source systems and collect market data among the one or more market data source systems. The data distribution system is also configured to store, in at least one database, one or more symbol elements linked to segments of the collected weather data and one or more pre-defined rules for generating weather symbology instructions having a pre-defined instruction structure. The data distribution system is further configured to generate the interactive GUI for display on the at least one user device, receive a weather data request from the at least one user device via the interactive GUI and determine a weather symbology instruction based on at least one requested symbol element indicated in the received weather data request and in accordance with the one or more pre-defined rules. The data distribution system is further configured to create at least one weather forecast dataset from among the collected weather data based on the weather symbology instruction, generate a presentation package comprising the weather forecast dataset and a portion of the collected market data such that the weather forecast dataset is integrated with said portion of the collected market data, present the presentation package on the interactive GUI and update the presentation package on the interactive GUI concurrent with changes to one or more of the weather data, the market data and user input via the at least one user device.

[0007] Aspects of the present disclosure also relate to data integration and distribution systems, methods and non-transitory computer readable medium. A system includes one or more weather data source systems configured to generate one or more weather data streams, one or more market data source systems configured to generate one or more market data streams and a data distribution system in communication with the one or more weather data source systems and the one or more market data source systems via at least one network. The data distribution system is configured to

store, in at least one database, a weather symbology comprising one or more pre-defined symbol elements linked to one or more segments of the one or more weather data streams, receive the one or more weather data streams among the one or more weather data source systems via at least one data feed and receive the one or more market data streams among the one or more market data source systems via the at least one data feed. The data distribution system is also configured to segment the received one or more weather data streams based on the one or more pre-defined symbol elements to form at least one segmented weather data stream, receive a weather data request from at least one user device via the at least one network, identify at least one symbol element among the one or pre-defined symbol elements indicated in the received weather data request, extract one or more segments of at least one segmented weather data stream based on the at least one identified symbol element to form at least one extracted weather stream, create a real-time weather dataset from the at least one extracted weather stream based on the weather data request and generate an integrated presentation package comprising the real-time weather forecast dataset and the received one or more market data streams. The data distribution system is further configured to at least one of expose the integrated presentation package to at least one application programming interface (API) and present the presentation package on a display of the at least one user device, and update the integrated presentation package concurrent with changes to one or more of the one or more weather data streams and the one or more market data.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0008] FIG. 1 is a functional block diagram of an example data distribution environment, according to an aspect of the present disclosure.

[0009] FIG. 2 is a functional block diagram of an example weather integration server, according to an aspect of the present disclosure.

[0010] FIG. 3A is a flowchart diagram of an example method of generating integrated weather and market data, according to an aspect of the present disclosure.

[0011] FIG. 3B is a flowchart diagram of an example method of generating integrated weather and market data for streaming data, according to another aspect of the present disclosure.

[0012] FIG. 3C is a hierarchical diagram illustrating an example weather symbology and use of the weather symbology to form various presentation packages, according to an aspect of the present disclosure.

[0013] FIG. 4A is screenshot including a chart of example weather forecast data and price data as a function of exchange time, according to an aspect of the present disclosure.

[0014] FIGS. 4B and 4C are screenshots of an example 16 day weather forecast for a particular location as a function of exchange time with respect to natural gas price data, according to an aspect of the present disclosure.

[0015] FIG. 4D is a screenshot of an example weather alpha workflow chart as a function of exchange time, according to an aspect of the present disclosure.

[0016] FIG. 5 is a screenshot including an example guidance chart associated with a guidance weather perspective, the guidance chart illustrating weather forecast data as a function of a forward time period, according to an aspect of the present disclosure.

[0017] FIG. 6A is a screenshot including an example progression chart associated with a progression weather perspective, according to an aspect of the present disclosure.

[0018] FIG. 6B is a screenshot of an example progression chart illustrating natural gas storage estimates and weather model progression as a function of forecast date, according to an aspect of the present disclosure.

[0019] FIG. 6C is an example progression chart for several weather models as a function of

forecast date, according to an aspect of the present disclosure.

[0020] FIG. 7A is an example initiation screen for generating an accuracy chart, according to an aspect of the present disclosure.

[0021] FIG. 7B is an example accuracy chart associated with an accuracy weather perspective, according to an aspect of the present disclosure.

[0022] FIG. 8A is an example user interface illustrating examples of symbology construction, according to an aspect of the present disclosure.

[0023] FIG. 8B is an example user interface illustrating examples of various weather symbology structures, according to an aspect of the present disclosure.

[0024] FIGS. 9A, 9B, 9C, 9D, 9E, 9F, 9G and 9H are screenshots of an example interactive GUI for providing interactive tools for viewing and interacting with integrated weather and market data, according to an aspect of the present disclosure.

[0025] FIGS. 10A, 10B, 10C, 10D, 10E, 10F, 10G, 10H, 10I, 10J and 10K are screenshots of an example interactive GUI for providing interactive tools for viewing and interacting with integrated weather and market data, according to another aspect of the present disclosure.

[0026] FIG. 11 is a functional block diagram of an example computer system according to an aspect of the present disclosure.

DETAILED DESCRIPTION

[0027] As discussed above, problems exist in the integration and distribution of disparate data types, including in a fully-automated (or near fully-automated) manner. One such combination of disparate data types involves weather data and market data. Each of these disparate data types may include historical data values, current (e.g., streaming/live) data values and future (e.g., forecast) data values. Each of these disparate data types also rely on underlying data and information that may be rapidly changing. Yet further, each of these disparate data types involve large volumes of data. For example, market data typically involves tens, if not hundreds, of thousands of live indications available to display at any given time, and where changes to market data may occur hundreds, if not thousands of times in a second. Similarly, weather data may involve continually changing weather observations from among a large number of locations (e.g., on a global scale, on a country-level scale, on a state-level scale, etc.), as well as weather forecast model data that may include a number of different models, and which models may be updated at different update intervals. All of the above variables make it technically difficult to manage data distribution and interaction for real-time distribution. Because weather and market data are also disparate data types, it becomes still further technically difficult to integrate both data types in an intelligent and meaningful manner in an interactive display, to provide insights into both data types (e.g., any correlations, predictive effects of one data type on the other data type and the like).

[0028] Market data, such as market price data may be accessed in watchlists, data tables, data grids, spreadsheets (e.g., Microsoft® Excel® via (real time data (RTD) links embedded into Excel), as well as interactive charts. These are basic functions of many conventional trading desktops (e.g., trading desktops provided by Bloomberg™, Refinitiv™ (e.g., Eikon™), Global View, DTN Exchange® and others) within the electronic financial trading community. These trading desktops were born in the 1980's and continue to evolve into highly specialized tools, designed to instantly display market data (using a time series server based technology) which was optimized for data delivery speed for viewing and interacting with market price data. Some conventional systems support plotting real-time market prices in real-time with updating charting tools.

[0029] Weather data has also experienced a similar evolution. Conventional interactions with weather data has been primarily using weather maps, considering it was the most efficient manner to view large amounts of data using a single image. Over the years, processing became more powerful and bandwidth became more plentiful, and both processing and bandwidth have become more affordable. The weather forecast models themselves became significantly larger in file size and the speed to process these larger weather forecast files, into maps, for internet based delivery

(as opposed to fax) became the focus. At the same time, the increased computational power also allowed for the extraction of raw weather forecast 'data' from these computer-based files, could quickly convert the forecast files (e.g., in a native file format such as gridded binary format (GRIB) files) and deliver city or zip code based weather forecasts in a few minutes. From this raw 'data', the creation of timely weather forecast charting tools to display a specific city's weather forecast, over the next few days, became popular, for example, on webpages and eventually cellphones.

[0030] The present disclosure is the first to combine these two existing charting capabilities into a unique weather charting and analysis tool that integrates both weather and market data (i.e., disparate data types), including in some examples both streaming weather data and streaming market data. In some examples, time series server based technology inside a trading desktop may be leveraged and used to uniquely create a one of a kind weather "weather symbology." The specialized symbology allows weather forecasts to be plotted "in exchange time", meaning the specific time when that weather forecast "element" was made available into the public domain (e.g., a forecasted temperature, wind, humidity, etc. and/or a value-added calculation which may convert forecasted temperature into a value representing forecasted natural gas consumption called a Gas Weighted Degrec Day (GWDD), etc.) as opposed to an "Interval start" time which displays the same forecast "element" (e.g., a predetermined number of days, such as 16 days forward). Importantly, the system of the present disclosure may contain pre-defined information in the symbology such as a weather model name, weather location, a "Forecast Valid Date (or Forecast Valid Day), Time of Day" and model hour in the future. The unique capability created by the present disclosure and offered inside a trading desktop of the present disclosure is the ability to instantly and easily visualize "Weather Alpha", which is the weather's financial impact on various traded market prices. For example, if a weather forecast for say "Forecast Valid Day 9" was significantly colder than the previous weather forecast and was released into the public domain at 10:31:12 am, which that streaming information (in some examples) then caused the price of Natural Gas, traded on an electronic Exchange, to move 5 cents in the following 10 seconds (after the release of this weather news/forecast data), users of a trading desktop of the present disclosure (also referred to herein as ICE Connect or ICE Connect's Trading Desktop) would have valuable insight over weather users, who simply review weather maps, or interact with weather data via some disparate source, which is not integrated with electronically traded market prices.

[0031] While this process of matching the "exchange time" (aka data receipt times) of both weather forecast data and market price data may appear simple on the surface, there are many complexities. The problems arise because conventional trading desktops were originally designed to handle prices for stocks, which involves plotting the market price with an "exchange time" (aka receipt time). Futures and forward contracts were subsequently incorporated and specialty views were created to accommodate an instrument which has a current market price in "exchange time" (aka receipt time) but also represents some point in the future. So a new symbol structure was created to represent these futures/forward contracts which represent future times the traded contract will expire, (such as a January natural gas contract, February, natural gas, etc.). These monthly futures contracts can also be stitched together to see a "forward curve" view (e.g., the January contract+February contract+March+ etc.) and all those individual market prices may be updated in real-time respectively.

[0032] The present disclosure handles weather data in a similar manner; in which a 16 day weather forecast (for example) is formed similar to a "forward curve." Each weather forecast may be stitched together (as it updates multiple times per day) thus creating a "forward curve" of a "forward curve." In some examples, the present disclosure allows users to interact with this weather data in one or more (e.g., three) primary user-weather perspective(s) to help organize the potentially multiple aspects of time related to the (e.g., streaming) data). The weather perspective(s) represent a type of (software) workflow that may provide access (e.g., visualization, observation and/or interaction tools) to weather data from different perspectives (e.g., slices of a

database of weather data), thereby enhancing the symbology structure and respective user-experience. In some aspects, the present disclosure may create a particular weather symbology to provide weather visualization/observation/interaction tools including one or more weather perspectives (e.g., three), and a weather alpha workflow that combines weather and market data. Significantly, such a weather alpha workflow has not been accomplished prior to the present disclosure. Advantages of the present disclosure include an integrated presentation package (including in some examples one or more weather forecast charts, tables, grids (e.g. where a user may select a particular area of the grid and a predetermined (e.g., streaming) view of weather and market data may be generated responsive to the selection), etc. integrated (e.g., overlaid) with market data). In some examples, the weather perspectives may be configured to provide more efficient access to weather data in an optimal manner, resulting in user interfaces that quickly render any suitable chart, table, analytic value extremely fast using static, periodic, aperiodic and/or streaming data.

[0033] In some examples, aspects of the present disclosure relate to weather integration systems and interactive GUIs that improve streaming weather data in an efficient manner into an electronic trading desktop (as well as other client applications). Weather integration systems of the present disclosure are unique in that the functionality of streaming weather forecast data into a trading desktop and combining (integrating) the weather data with streaming market data (e.g., streaming price data) to create actionable knowledge does not exist outside of the present disclosure. The weather integration system is able to provide more credible weather forecast models that may be made available from a greater number of weather data sources. The weather integration system may also provide forecasts that extend farther out in time and may be configured to update the weather forecast data with higher temporal and spatial resolution creating larger and larger file sizes, It is understood that the ability to handle and process an ever growing amount of data is a technical problem, thus the ability to handle and process streaming raw data, and convert this streaming data into actionable knowledge via an interactive GUI (as performed by the systems of the present disclosure) represent a technical solution to a technical problem.

[0034] In some examples, weather integration systems and interactive GUIs of the present disclosure provides a unique technique of integrating data in an exchange time to line up changing perceptions of weather forecast model(s) with changes in market prices. The weather integration system creates a unique weather symbology that is optimally designed to blend with other data streams which are configured to operate within a time series server. In general, the weather symbology may be configured for both weather data streams and non-weather (e.g., market data) data streams. The weather integration system of the present disclosure may be configured to provide access to raw data via optimally arranged dataset(s) in a manner that may be helpful for indicating an impact of weather data on market data, such as providing user interfaces for back-testing changing market prices as they are associated with changing weather perspectives. In some examples weather data (e.g., streaming and/or static data) may utilize the novel symbology of the present disclosure, which allows weather data (live and/or historical data) to flow seamlessly through a trading desktop (e.g., for interactive presentation on a GUI), as well as for providing weather data to external systems (e.g., for further analysis, visualization, etc.). The symbology of the present disclosure is unique in that it employs the smallest possible building blocks to organize weather data in nearly any possible (user-customizable) manner. The highly granular symbology of the present disclosure allows users of a trading desktop to review weather data in a manner similar to market price data as well as combine any element of weather with any granular element of market price data.

[0035] The weather integration systems and interactive GUIs of the present disclosure provide users with unique advantages, including indicating precisely how changing weather perspectives may directly influence market prices. The weather integration system may provide various weather maps, various weather data and various value-analytics.

[0036] The weather maps may include, in some examples, weather maps as part of a robust weather maps application which may stream thousands of maps for each forecast hour from various government entity weather models being released into the public domain. The weather maps provided by the weather integration system may be provided very rapidly, and may be configured to provide comprehensive information in a manner that is easy to use (e.g., user-friendly). Weather data provided by the weather integration system may be provided quickly and in a manner to be easily assessable. In some examples, the weather data may include, without being limited to, one or more of a forecast confidence (risk), a forecast volatility, a forecast accuracy; a forecast alpha and a rich history of historical forecasts and/or observations. The weather integration system may provide value-added analytics be configured to instantly convert weather data into accurate and actionable knowledge, tailored specifically for traded instruments, including, in a non-limiting example, to one or more of Gas Weighted Degree Day (GWDD), Electricity Weighted Degree Day, Propane Weighted Degree Day, Heating Oil Weighted Degree Day and the like.

[0037] In some examples, the weather integration system of the present disclosure may be configured to convert extremely large streams of weather and price data into actionable knowledge, including in a manner that is faster than conventional systems, and provides improved technical features including improved data speed, reliability, comprehensive data, a unique weather symbology that does not exist outside of the present disclosure and unique transformative weather workflows.

[0038] In some examples, the weather integration system may be configured to provide a data speed to ingest, process and deliver global government weather forecast (e.g., raw) data, to users, in both data and map formats with low latency. In some examples, the weather integration system may be configured to provide improved reliability of weather data, including the ability to acquire, render and transmit (raw) weather data in a low latency fashion into various types of weather delivery mechanisms. In some examples, the weather integration system may provide comprehensive weather data, by using high (including highest) resolution data and accurate (including most accurate) data versions, of the most critical global weather forecast models (which few (if any) other weather providers currently offer), as well as an enhanced temporal and spatial resolution.

[0039] In some examples, the weather integration system may provide a unique weather symbology having unique symbol elements (also referred to herein as “Weather Legos”). The Weather Legos may be configured to breakdown streaming weather data into the smallest possible building blocks to power a time series server-based trading desktop, allowing users to instantly and easily interact with both weather and price data (e.g., through time & space). In some examples, “Weather Legos” allow users to create any weather display imaginable, or easily replicate a favorite display from another source. In some examples, the weather integration system may provide transformative weather workflows, which workflows may provide an improved speed of understanding the impact of weather including on market data. The workflows may leverage the “Weather Legos” and improved data speed, together with the full power of an electronic trading desktop, to create unique displays that allow users to more quickly and more precisely understand how changes in perceived weather may directly influence market prices. The unique displays of the present disclosure transform how market prices interact with weather data in the financial community. The displays of the present disclosure allow users to convert more weather and price data into actionable knowledge, and in a manner that is faster than conventional systems.

[0040] The weather integration system of the present disclosure are also unique in that it fully integrates two unique workflows (weather and market data, an example of disparate data structures) that are conventionally handled in a completely separate manner, because conventional systems were unable to integrate the two streaming workflows due to the technical complexities of the weather and market workflows. Instead, at best, some conventional trading desktops may simply provide access to weather (e.g., via i-frames) but cannot provide integrated weather and market data (e.g., weather data overlaid with market data). The weather integration system of the

present disclosure (including the end-to-end infrastructure) is the first and only weather integration system of its kind. In some examples, the weather integration system may provide opportunities to scale electronic trading operations geared to leverage changing weather perspectives across many different asset classes, and may provide unique displays that improve identification of trading opportunities, including faster identification of trading opportunities.

[0041] In some examples, the weather integration system may derive one or more preliminary daily forecast variables (e.g., daily average temperature, gas weighted degree day (GWDD), heating degree day (HDD), cooling degree day (CDD), etc.), and may provide a “% Complete” value prior to the particular model's completion of that forecast day. In other words, each forecast day may have twenty four forecast model hours. When the noon forecast arrives, twelve of the twenty four possible forecast model files exist, in the public domain, so that the forecast day is 50% complete. When the 1pm forecast model data arrives, the forecast day is now 54% complete, and so on. This innovative approach provides users valuable preliminary daily forecast data which may be directionally accurate. In other words, users may be presented a forecast day before (e.g., approximately about 4 minutes before) other weather vendors post their daily forecast data (from the same government weather forecast model).

[0042] In some examples, the weather integration system may provide an improved speed of understanding of weather and market data. The unique weather workflows (weather alpha and perspective workflows, described further below) allows users to make more informed and faster weather and price decisions. The same centralized weather workflows may easily allow a user to scale this unique process across an electronic trading floor and across electronically traded instruments, thereby allowing each trader to more easily monitor multiple traded markets, as it relates to changing weather perspectives (e.g., with greater accuracy and faster speed).

[0043] In some examples, the weather integration system may provide comprehensive model data, including higher resolution forecast data from government entity models. Because the weather integration system may provide a low latency infrastructure, the weather integration system, in some examples, may be configured to move large amounts of data around the world in seconds. This unique infrastructure may be leveraged to extract higher resolution versions of various forecast data including, in some examples, higher resolution versions of the GFS, GFS ENS, ECMWF and ECMWF ENS models. In some examples, the weather integration system may provide model data with enhanced spatial resolution (e.g., $\frac{1}{4}$ degree, whereas conventional resolution may be $\frac{1}{2}$ degree) and enhanced time (temporal) resolution. In some examples, enhanced spatial resolution may provide a slightly more accurate weather model version (e.g., picking up on micro climates) which lower resolution versions may not be able to provide. In some examples, enhanced temporal resolution may include hourly time steps, meaning that users may receive forecast updates, for example, every 20 seconds. In contrast, conventional temporal resolution may be about three hourly updates which may occur every 60 seconds. In addition, in some examples, twenty four model hours (e.g., time steps) may be included in the first five days (120 hrs) of the GFS weather forecast model.

[0044] In some examples, the weather integration system may leverage the unique weather symbology combined with streaming market price data (e.g., market data stream(s) together with a trading desktop functionality to create an almost limitless combination of symbol element (“Weather Legos”) designs to provide unique interactive GUIs, client applications, client services and trading desktop applications. In some examples, the weather symbology is configured to treat weather data like a traded instrument, so any weather symbol may be entered directly into a chart, table, watchlist, etc. . . . to view the weather data directly. In some examples, the weather symbology may be configured to operate in trading desktops, spreadsheet applications and with one or more object-oriented programming languages (e.g., Python™).

[0045] In some examples, weather data may be accessed in real-time using the symbol elements, and the weather integration system may be configured to leverage both streaming weather data and

streaming price data inside an electronic financial desktop. The unique weather symbol clements may also allow weather data to be accessed from one or more separate directions, including, in some examples three primary weather perspectives, including guidance, progression and (forecast) accuracy perspectives.

[0046] In some examples, the guidance perspective may be configured to assess weather forecast risk and confidence. For example, if all of the weather forecasts agree, there is less risk embedded into the market prices. Alternatively if there is a wide range in the forecasts, then there is greater risk (less confidence) in the perceived future weather, presenting greater price volatility of the market prices. In some examples, the progression perspective may be configured to assess a 'run-to-run' forecast volatility as well as how all weather forecast models relate to each other. In some examples, the progression perspective may use the principles of convergence and mean reversion to help users spot short-term and longer-term electronic trading opportunities. In some examples, the accuracy perspective may be suitable for users who take (or make) physical delivery of weather-related instruments such as power, natural gas, natural gas liquids and the like. For example, often there is a disconnect between the perception of the weather, at the time a monthly futures contract expires, versus what weather will actually occur inside of that month. It is important to understand the probability a weather forecast will be accurate (or inaccurate). The accuracy perspective may aid in such an understanding.

[0047] Referring now to FIG. 1, a functional block diagram of an example data distribution environment **100** (environment **100** herein) is shown. Environment **100** may include data integration/distribution (DID) system **102**, one or more weather data sources **104**, one or more market data sources **106** and one or more user devices **108** (associated with at least one user). In some examples, one or more components of environment **100** (e.g., DID system **102**, weather data source(s) **104**, market data source(s) **106** and/or user device(s) **108**) may be communicatively coupled via one or more networks **110**. Network(s) **110** may include, for example, a private network (e.g., a local area network (LAN), a wide area network (WAN), intranet, etc.) and/or a public network (e.g., the Internet).

[0048] Weather data source(s) **104** may include, without being limited to, weather forecast model sources, one or more weather observations data sources associated with one or more locations and the like. In some examples, sources for weather forecast model(s) may provide weather model data for generating weather forecasts for one or more geographical regions. In one non-limiting example, the weather forecast model source(s) may include one or more of National Oceanic and Atmospheric Administration (NOAA) United States (US) Global Forecast System (GFS), NOAA GFS global ensemble (GFS ENS), NOAA GFS global ensemble Extension (GFS ENS EXT), NOAA's Climate Forecast System Ensemble model (CFS ENS), European Center for Medium-Range Weather Forecast (ECMWF), ECMWF Ensemble, ECMWF Ensemble Extension (ECMWF EXT), ECMWF Seasonal model (SEAS) and its corresponding extension model (SEAS EXT) and other gridded binary format (GRIB) weather forecast model source(s). In general, a weather forecast model source may run a weather forecast model (e.g., periodically such as quarterly, monthly, multiple times per week, one or more times per day, asynchronously, in response to a predetermined condition, etc.) and may provide weather forecast data for up to a predetermined number of days. For example, the model for the GFS is run four times a day and may produce forecasts for up to 16 days in advance and run, for example, four times per day. As another example, the CFS ENS may run daily and may forecast out nine months into the future. In one non-limiting example, source(s) for weather observations data may include one or more of Meteorological Aerodrome Report (METAR) data and Climate Forecast System (CFS) data including CFS Reanalysis (CFS-R) data for one or more locations (e.g., one or more airports, weather stations, etc.). In some examples, CFS-R data may be used to provide global observation estimates, for an even greater number of weather parameters, as estimates which METAR sensors and/or limited physical locations may not provide. In general, weather observations data may be

updated periodically (e.g., hourly, daily, monthly, quarterly, etc.), and may include various weather data (e.g., temperature, dew point, wind direction, wind speed, precipitation, cloud cover, visibility, barometric pressure and the like, as well as derived values such as GWDD) for one or more weather observation locations. In general, weather data source(s) **104** may include any suitable weather data sources for weather forecast model(s) and weather observations data associated with one or more locations. In general, weather data source(s) **104** that may be in communication with DID system **102** may comprise a server computer, a desktop computer, a laptop, a smartphone, or any other electronic device known in the art configured to capture data, receive data, generate weather forecast model data, store data and/or disseminate any suitable weather data.

[0049] Market data source(s) **106** may include, in general, any electronic source for market data (e.g. price data and/or trade-related data associated with one or more tradeable financial instruments). Non-limiting examples of types of market data that may be provided by market data source(s) **106** may include one or more of equities data, derivatives data, data associated with one or more indices, market depth data, energy market data, fundamental data, inter-dealer broker (IDB) data and over-the-counter (OTC) market data. In some examples, market data source(s) **106** may also include one or more news sources that may provide information associated with market data (e.g., observations on market data trends, news stories that may affect market data and the like). In some examples, fundamental data may include one or more sources which contain structured and/or non-structured data which may provide information associated with market data impacting supply, demand and/or transportation (such as shipping data, electricity emissions, natural gas pipeline flows, etc.). In general, market data source(s) **106** that may be in communication with DID system **102** may comprise a server computer, a desktop computer, a laptop, a smartphone, or any other electronic device known in the art configured to capture data, receive data, store data and/or disseminate any suitable market data.

[0050] User device(s) **108** may include, without limit, any combination of mobile and/or stationary communication device such as mobile phones, smart phones, tablets computers, laptop computers, desktop computers, server computers or any other computing device configured to capture, receive, store and/or disseminate any suitable data. In some examples, user device(s) **108** may include one or more external computer systems configured to receive weather and/or market data as processed by DID system **102**, and may provide additional analysis, user interaction and/or distribution functionality. In one embodiment, user device(s) **108** may include a non-transitory memory, one or more processors including machine readable instructions, a communications interface which may be used to communicate with DID system **102**, a user input interface for inputting data and/or information to user device(s) **108** and/or a user display interface for presenting data and/or information on user device(s). In some examples, the user input interface and the user display interface may be configured as a graphical user interface (GUI) (such as interactive GUI **900** shown in FIGS. **9A-9H**). User device(s) **108** may also be configured to provide interactive GUI **120** of DID system **102** on the GUI of user device(s) **108**. In some examples, user device(s) **108** may include computer system **1100** (shown in FIG. **11**).

[0051] DID system **102** may include data collector **112**, data feed distributor **114**, weather integration server **116**, time series server **118**, interactive GUI **120**, one or more data caches **124**, one or more client applications and/or services **126**, one or more data caches **124**, and client/server connection manager **128**. In some examples, one or more components **112-128** of DID system **102** may communicate with each other via a data and control bus (not shown). Although DID system **102** is shown in FIG. **1** as one component (e.g., a server), DID system **102** may include one or more components (e.g., one or more servers), whether co-located or linked across one or more networks.

[0052] Although not shown, DID system **102** may include at least one processor (e.g., processing device **1102** shown in FIG. **11**) and non-transitory memory (e.g., memory **1106** shown in FIG. **11**) storing one or more routines and or algorithms for performing the functions of DID system **102**

described herein. An example implementation of one or more components of DID system **102** is shown by computer system **1100** (shown in FIG. **11**).

[0053] In some examples, DID system **102** may obtain weather data and market data from among respective weather data source(s) **104** and market data source(s) **106** via data collector **112**. Although not shown, data collector may include one or more input and/or output interfaces (e.g., an electronic device including hardware circuitry, an application on an electronic device) for communication with other components of environment **100** (e.g., weather data source(s) **104** and market data source(s) **106** (and, in some examples user device(s) **108**)).

[0054] Data collector **112** may be configured to obtain weather and market data through any suitable electronic data collection technique. For example, data collector **112** may obtain data through or more data feeds (including streaming (live) data feed(s)), one or more file transfers (including, in some examples, one or more secure file transfers), by data being pushed to DID system **102** and/or by DID system **102** pulling and/or extracting data and/or information from among weather data source(s) **104** and/or market data source(s) **106**. In some examples, data collector **112** may include security protection (e.g., encryption, decryption) to protect the integrity of the obtained data.

[0055] In some examples, data collector **112** may perform one or more of filtering, data normalization, data reformatting, data aggregation and the like. In some examples, electronic data that may be obtained by data collector **112** may include data that may be unable to be processed, data that includes one or more errors and the like. In some examples, different sources among weather data source(s) **104** and market data source(s) **106** may transmit data with various unique, non-standard values and/or data formats (e.g., proprietary formats). Furthermore, data content may correspond to different forms of data, such as different currencies, date formats, time periods, etc.

[0056] Non-limiting examples of data filtering that may be performed by data collector **112** may include excluding null data values, excluding corrupted data, excluding outlier data values (e.g., a data value outside of a pre-defined value range, outside of a pre-defined time range) and the like. In some examples, data collector **112** may reformat the collected data to one or more common formats and/or normalize the collected data. In some examples, data collector **112** may be configured to aggregate and/or combine at least a portion of the collected data (e.g., according to one or more pre-defined categories (such as a type of data, one or more predetermined time periods, etc.)).

[0057] Data feed distributor **114** may be configured to receive the collected data from data collector **112**. Data feed distributor **114** may also be configured to communicate with weather integration server **116** and time series server **118**. In some examples, data feed distributor **114** may also be configured to communicate with client application(s)/service(s) **126**. In some examples, data feed distributor **114** may also be configured to communicate with data cache(s) **124**.

[0058] In general, data feed distributor **114** may be configured to process the received data (e.g., the collected market and weather data), and may distribute the received data to one or more among weather integration server **116**, time series server **118**, data cache(s) **124** and client application(s)/service(s) **126** (not all connections with data feed distributor **114** shown). Data feed distributor **114** may determine which component to distribute the received data to, based on one or more characteristics of the received data (e.g., data type, one or more predetermined conditions, one or more predetermined parameters and the like). In general, data feed distributor **114** may distribute weather data among the received data (collected from weather data source(s) **104**) to weather integration server **116**, and may distribute market data among the received data (collected from market data source(s) **106**) to time series server **118**.

[0059] Weather integration server **116** may be configured to receive weather data from data feed distributor **114**, including weather forecast model data from among one or more weather forecast data models and weather observation data from one or more locations. Weather integration server **116** may be configured to communicate with time series server **118** and interactive GUI **120**.

Weather integration server **116** may be configured to segment (e.g., separate and/or index) the

received weather data (from among the weather observations data and the weather forecast model data) into one or more segments in accordance with a predetermined weather symbology. (Weather symbology is described further below). In some examples, weather integration server **116** may also store at least a portion of the segmented weather data. Weather integration server **116** may provide the segmented weather data to time series server **118**. In some examples, weather integration server **116** may also generate (and store) one or more weather maps from the weather forecast model data. In some examples, weather integration server **116** may provide weather map(s) to interactive GUI **120**. In general, weather integration server **116** may provide segmented weather data to time series server **118** (as well as weather map(s) to interactive GUI **120**) that may include real-time (e.g., streaming) data, near-real-time data, historical (e.g., static data) and/or any combination thereof. [0060] In some examples, the segmented weather data may be generated to provide (weather symbology-segmented) weather data to time series server **118** as a function of exchange time and/or interval time. Importantly, the segmented weather data, as generated by weather integration server **116**, is in a format that can be readily integrated with market data within time series server **118**. In some examples, the segmented weather data may be automatically updated based on (i.e., updated concurrent with) changes to the received weather data (e.g., new observation data, any updates to the weather forecast model data, etc.).

[0061] Time series server **118** may be configured to receive market data from data feed distributor **114** and segmented weather data from weather integration server **116**. Time series server **118** may also receive at least one weather data request (e.g., via interactive GUI **120**, via client application(s)/service(s) **126**). In some examples, the weather data request may be associated with at least one symbol element, and may be converted to weather symbology instruction. Time series server **118** may extract one or more portions of the segmented weather data (received from weather integration server **116**) in accordance with the weather symbology instruction. In general, the weather symbology instruction may include one or more symbol elements (such as forecast location, weather perspective, weather model). In some examples, the weather symbology instruction may also include one or more variables (e.g., field(s) and/or condition(s)). (Examples of weather symbology instructions and weather perspectives are described further below.) Time series server **118** may be further configured to integrate the extracted weather data (in accordance with the weather symbology instruction) with at least a portion of market data (e.g., received from data distributor **114**) to form an integrated presentation package. The integrated presentation package may be provided to interactive GUI **120** and/or client application(s)/service(s) **126** (including, in some examples, at least one application programming interface (API)).

[0062] In some examples, time series server **118** may include a tick engine and at least one database, for processing large amounts of market data that may change rapidly (e.g., user orders, market events, quotes, etc.). Time series server **118** may also include a presentation tool for generating presentation packages, including integrated weather and market data presentation packages. For example, time series server **118** may combine the extracted weather data (including, for example, weather forecasts, weather observations, weather symbology indications) with at least a portion of the market data (e.g., based on user input, pre-defined parameters, etc.) to form at least one integrated presentation package suitable for presentation (and interaction) (e.g., on interaction GUI **120**, via an API, etc.). Although examples are described where presentation package includes both weather and market data, in some examples, at least one presentation package may be generated that includes (packaged) extracted weather data.

[0063] Although not shown, time series server **118** may include a controller (e.g., at least one processor, a microcontroller and the like, such as processing device **1102** shown in FIG. **11**) and non-transitory memory (e.g., memory **1106** shown in FIG. **11**) storing one or more routines and/or algorithms for performing the functions of time series server **118** described herein. An example implementation of one or more components of time series server **118** is shown by computer system **1100** (shown in FIG. **11**).

[0064] The integrated presentation package may include any suitable data and/or information (including weather data, market data, other suitable data and/or any combination thereof) configured in a manner for presentation and/or user interaction, including, but not limited to, one or more of at least one chart, at least one map, at least one watchlist, at least one alert, at least one grid application, at least one table, user input options including weather symbology input options and the like. In some examples, at least a portion of the data and/or information presented in the integrated presentation package may be user-customizable (including via the use of weather symbology). In some examples, the data and/or information in the presentation package may include one or more (live) data streams, such that the data and/or information presented by the presentation package may be updated concurrently with the data in the data stream(s). In some examples, the presentation package may also include historical (e.g., static data). In some examples, the presentation package may include at least one interactive weather grid that may be configured to allow user selection of a portion of the grid (e.g., clicking on a particular area with a pointing device). Responsive to the user selection, interactive GUI **120** may generate a predetermined streaming view of weather and market data associated with the selected portion.

[0065] An example of weather integration server **116** together with time series server **118** is described further below with respect to FIG. 2.

[0066] Interactive GUI **120** may be configured to generate weather impact dashboard **122** having a uniquely configured arrangement of one or more input regions, one or more notification regions and one or more display regions. In some examples, one or more portions of weather impact dashboard **122** may be automatically updated (including in real-time such as streaming data or near real-time) responsive to changes in weather data and/or market data. In some examples, the display region(s) may include an interactive display that permits/prompts user input and may be automatically updated in response to user input.

[0067] In some examples, weather impact dashboard **122** may include different configurations based on an underlying application and/or service (e.g., among client application(s)/service(s) **126** for which it is launched. For example, weather impact dashboard **122** may have a first configuration for a mobile application, may have a second (different) configuration for a desktop application and may have a third (different) configuration for a spreadsheet application. In some examples, weather impact dashboard **122** may include one or more windows with different configurations depending upon a particular weather perspective. In some examples, weather impact dashboard **122** may provide user input of a weather symbology instruction directly (e.g., a weather data request may include a weather symbology instruction itself). In some examples, weather impact dashboard **122** may provide one or more user input prompts that may be converted to a weather symbology instruction.

[0068] Examples of interactive GUI **120** are described further below with respect to FIGS. 9A-10F.

[0069] In some embodiments, DID system **102** may include data cache(s) **124** (e.g., a hardware component and/or software component) for storing data and/or information associated with the various functions of DID system **102**. Data cache(s) **124** may be configured to store collected data (e.g., by data collector **112** and distributed by data feed distributor **114**), including, for example, weather data from among weather data source(s) **104** and market data from among market data source(s) **106**. In some examples, data cache(s) **124** may store requested weather datasets (generated by weather integration server **116**) and/or integrated presentation package(s) (generated by time series server **118**). In general, data cache(s) **124** may be configured to store any data in which faster data retrieval of the data for future requests may be useful. In some examples, data cache(s) **124** may include a data manager (DM) (not shown) for controlling collection and/or retrieval of data in data cache(s) **124**.

[0070] In some examples, DID system **102** may include storage (e.g., one or more databases) to store any parameters, configurations, functions and/or other suitable data and/or information that may be useful for one or more of components **112-128**. In some examples, the storage may be

configured to store data distributed by data feed distributor **114** and/or data collected by data collector **112**.

[0071] In some examples, DID system **102** may include client application(s)/service(s) **126** for creating instances of application(s) and/or service(s) for distributing data to user device(s) **108**. In one non-limiting example, client application(s)/service(s) **126** may include one or more of a trading desktop, a spreadsheet application, a mobile application (e.g., an application that may be suitable for display screen of a mobile device such as a smartphone) streaming (e.g. raw, normalized, filtered and/or aggregated) data feeds and/or one or more APIs for receiving data in one or more object-oriented programming languages (e.g., Python™, R, etc.). In some examples, client application(s)/service(s) **126** may create instance(s) of application(s) and/or service(s) including interactive GUI **120** having at least one configuration of weather impact dashboard **122**. In some examples, DID system **102** may be configured to expose a presentation package to one or more API(s) among client application(s)/service(s) **126**.

[0072] In some examples, DID system **102** may include a client/server connection manager **128** (also referred to as connection manager **128**) configured to manage client to server connection requests. In general, connection manager **128** may manage connection request(s) of user device(s) **108** to DID system **102**, to access client application(s)/service(s) **126**.

[0073] Referring next to FIG. 2, a functional block diagram of example weather integration server **116** is shown. Weather integration server **116** may include observations collector **204**, one or more databases **206**, forecast/observation generator **210**, map generator **212**, weather symbology storage **214**, storage **216** for one or more weather structures and one or more databases **220**. In some examples, one or more components **204-216** and **220** of weather integration server **116** may communicate with each other via a data and control bus (not shown).

[0074] Although not shown, weather integration server **116** may include a controller (e.g., at least one processor, a microcontroller and the like, such as processing device **1102** shown in FIG. 11) and non-transitory memory (e.g., memory **1106** shown in FIG. 11) storing one or more routines and or algorithms for performing the functions of weather integration server **116** described herein. An example implementation of one or more components of weather integration server **116** is shown by computer system **1100** (shown in FIG. 11).

[0075] Observations collector **204** may be configured to obtain weather observations data **202** associated with one or more weather observation locations (e.g., from among weather data source(s) **104**). In a non-limiting example, weather observations data **202** may include METAR, CFS-R observations data and CFS-R generated climatological data (e.g., a trailing 10 year climatology, a trailing 15 year climatology, a trailing 30 yr Climatology, etc.). In some examples, observations collector **204** may be configured to separate METAR and CFS-R observations data, CFS-R climatology data, and may process each type of data separately. In some examples, observations collector **204** may be configured to parse weather observations data **202**, to identify one or more elements of weather observation data **202** that may be useful for generating segmented weather data **218**. In some examples, observations collector **204** may generate one or more identifiers for the identified elements of the (parsed) weather observations data **202**. In some examples, the identifier(s) may include an observation date, a time as to when the weather observation was observed and/or a time-stamp indicating when a particular value/set of values was received.

[0076] Observations collector **204** may also store observations data **202** (after parsing) in database(s) **206** (e.g., as historical observations). Database(s) **206** may store observations data in any suitable format/arrangement for efficient data storage and retrieval. In some examples, database(s) **206** may store (parsed) weather observations data **202** based on identifiers generated by observations collector **204**. Observations collector **204** may also provide at least a portion of weather observations data **202** (e.g., after parsing and/or from database(s) **206** to forecast/observation generator **210**.

[0077] Forecast/observation generator **210** may be configured to obtain weather forecast model data **208** (e.g., from among weather data source(s) **104**). In a non-limiting example, weather forecast model data **208** may include weather forecast models from GFS, GFS ENS, GFS ENS EXT, CFS ENS, ECMWF, ECMWF ENS, ECMWF ENS EXT, SAEAS and SEAS EXT forecast model sources. In one example, one or more (in some examples all) forecast models may originate in a GRIB file format and may be instantly converted into city level data for forecast valid times for a number of weather forecast variables, which may then be immediately run through time series server **118** so the data can stream through DID system **102**. Forecast/observation generator **210** may also obtain weather observations data **202** from observations collector **204** (and/or from database(s) **206**). Forecast/observation generator **210** may also be configured to communicate with weather symbology storage **214** and weather structure(s) storage **216**. In general, forecast/observation generator **210** may be configured to use pre-defined symbol elements (stored in weather symbology storage **214**) to segment forecast model data **208** and/or observations data **202** (e.g., via observations collector **204** and/or stored observation(s) via database(s) **206**) into one or more pre-defined segments of weather data, to form segmented weather data **218**. In some examples, at least a portion of forecast model data **208** may be stored in one or more databases (not shown), and forecast/observation generator **210** may use at least a portion of previously stored forecast model data to generate segmented weather data **218**. Segmented weather data **218** may be provided to time series server **118**.

[0078] Weather symbology storage **214** may be configured to store a pre-defined weather symbology. The pre-defined weather symbology may include one or more pre-defined symbol elements, one or more pre-defined variables (e.g. pre-defined condition(s) and/or fields) and one or more pre-defined rules for combining and/or arranging symbol elements into at least one weather symbology instruction having a pre-defined instruction structure. The symbol elements may be linked to respective segments of the weather data (including, in some examples, segments of weather data streams). The weather data may be segmented, (by forecast/observation generator **210**) in accordance with the symbol elements of the weather symbology stored in storage **214**. An example weather symbology is shown in FIG. 3C. In some examples, a weather symbology instruction may be used to extract portions of segmented weather data (e.g., by time series server **118**) to create integrated presentation package **226**.

[0079] Weather structure(s) storage **216** may be configured to store one or more pre-defined weather structures having one or more respective pre-defined rules (e.g., software workflows) for generating one or more pre-defined weather alpha workflows and one or more pre-defined weather perspectives. Time series server **118** may use the pre-defined weather structure(s) to create integrated presentation package **226**. In general, the pre-defined weather perspective(s) may represent workflows (e.g., having one or more pre-defined rules) for converting segmented weather data **218** into user-customizable actionable knowledge (e.g., providing one or more different ways to assess various aspects of segmented weather data **218** that may not be directly observable by simply presenting weather forecast(s) and/or weather observation(s)). In a non-limiting example, the weather perspective(s) may include one or more pre-defined rules for generating integrated presentation package **226** in accordance with one or more of a guidance workflow, a progression workflow and an accuracy workflow. Examples of weather perspectives are described further below. In a non-limiting examples, the one or more pre-defined weather alpha workflows may include pre-defined rules for generating at least one of a weather macro alpha and a weather micro alpha (described further below).

[0080] In general, a weather symbology instruction (associated with weather data request **220**), may be formed (e.g., by time series server **118**) having the pre-defined instruction structure, using weather symbol element(s) and variable(s) in accordance the pre-defined rule(s) stored in weather symbology storage **214**. In some examples, the weather symbology instruction may include a weather structure (e.g., a perspective), an observation location and a weather forecast model. The

weather symbology instruction may include additional condition and/or function information, such as one or more of a model run, forecast valid days, forecast valid hours, etc. In some examples, the weather symbology instruction may be created based on one or more data requests **224** (e.g., received via user device(s) **108**). Predefined rule(s) associated with the weather structure (of the weather symbology instruction) may be obtained from weather structure(s) storage **216** and may be used to create integrated presentation package **226**. In some examples, integrated presentation package **226** may include at least a portion of segmented weather data **218** (e.g., based on a weather alpha workflow and/or a weather perspective) integrated together with market data **222** (received by time series server **118**). In some examples, integrated presentation package **226** (at least a portion) may include at least a portion of segmented weather data **218** (e.g., without any market data **222**).

[0081] Although weather symbology storage **214** and weather structure(s) storage **216** are shown as being part of weather integration server **116**, storage components **214** and/or **216** may be external to weather integration server **116**. As shown in FIG. 2, time series server **118** may be communicatively coupled to storage components **214** and **216**.

[0082] Map generator **212** may receive at least a portion of forecast model data **208**, and may render one or more weather forecast maps via any suitable well-known rendering technique. Map generator **212** may also store one or more weather forecast maps in database(s) **220**. Database(s) **220** may store weather forecast map(s) in any suitable format/arrangement for efficient data storage and retrieval. One or more weather maps may be provided to interactive GUI **120** (e.g., via database(s) **220** and/or via map generator **212**). Examples of types of maps that may be rendered are described further below with respect to FIG. 10C.

[0083] In operation, forecast/observation generator **210** may create segmented weather data of the collected (e.g., streaming) forecast and/or observations data) in accordance with the pre-defined symbol elements (and in some examples, symbology variables) stored in storage **214**. In some examples, the weather data may be segmented by indexing portions of the weather data to the (linked) symbol elements (and symbology variables in some examples) and/or separating the weather data into subsets linked to the symbol elements (and symbology variables in some examples). In some examples, at least a portion of segmented weather data **218** may also be stored in one or more databases (not shown). Time series server **118** may receive at least one (weather) data request **224** (e.g., from user device(s) **108** via interactive GUI **120**). Response to data request **224**, time series server **118** may convert data request(s) **224** to a corresponding weather symbology instruction, based on pre-defined weather symbology rule(s) in weather symbology storage **214** (or may receive the weather symbology instruction directly in data request **224**). Time series server **118** may determine which segment(s) of segmented weather data **218** to extract based on the symbol element(s) in the weather symbology instruction (that are linked to segment(s) of segmented weather data). Time series server **118** may also determine a particular weather structure among weather structure(s) stored in storage **216**, based on the weather symbology instruction. Time series server **118** may create integrated presentation package **226** in accordance with corresponding pre-defined weather structure rules (e.g., workflow) for the particular weather structure. In some examples, time series server **118** may stitch together a weather forecast in accordance with the weather symbology instruction, by extracting segment(s) of segmented weather data **218** according to the particular weather structure. Moreover, time series server **118** may generate integrated presentation package **226** as a function of multiple time series formats, including exchange time and interval time, such that extracted segment(s) of segmented weather data **218** may be easily integrated with market data **222** by time series server **118**, so that end users (e.g., via interactive GUI **120**, via client application(s)/service(s) **126**) may readily identify any impacts of weather data on market data **222**.

[0084] In some examples, integrated presentation package **226** may be generated based on predetermined parameters (e.g., determined by weather integration server **116**, determined by time

series server **118**, defined by the user, etc.), without receiving any data request **224** (and/or independent of any data request **224**). For example, an initial launch page on weather impact dashboard **122** of interactive GUI **120** may be generated based on the predetermined parameters (via extracting portions of segmented weather data **218** in accordance with the predetermined parameters). A user of user device(s) **108** may then modify and/or customize integrated presentation package **226** that is ultimately presented on weather impact dashboard **122** by data request(s) **224**.

Weather Workflows Overview

[0085] Before describing details of the symbology structure, an overview of example weather workflows (e.g., weather alpha and weather perspectives) of the present disclosure for interacting with weather data is provided. The weather alpha and weather perspectives workflows are unique to the present disclosure and have never been integrated into a trading desktop or any system (including any streaming system) containing market prices prior to the present disclosure. Although the present disclosure describes three primary weather perspective workflows, it is understood that the present disclosure is not limited to the example workflows described below, and that any suitable weather workflow for interacting with weather and market data to provide meaningful indications of weather impact on market data are also within the scope of the present disclosure. In some examples, the weather perspective and weather alpha workflows may be used by any suitable end-user including, without being limited to, traders, analysts, meteorologists, trading floor quants, investors, hedgers and the like.

[0086] Weather Alpha: A system which allows for a data display or charting display of weather forecast variables (like temperature, Degree Day, GWDD, wind, humidity, etc. . . .) from a specific weather forecast model or multiple weather forecast models, initialized and released at a specific time of day (model run) from which a specific location or locations (e.g., city or region) which may (or may not) include preliminary estimated forecast values such as ‘% complete’ or ‘Fast& Full’ or partial week month or seasonal aggregations called “Progress’ or pre-calculated (streaming) values (i.e. “Change” from previous, “Anomaly/Difference” from Climatology) to be displayed in real-time, alongside market prices for any traded instrument which includes but is not limited to one or more traded commodities, equity or bond instruments which may or may not include futures, forward and/or various types of option contracts. The ‘weather symbology’ of the present disclosure allows the user a method to easily access weather forecast data in at least one of a “Seasonal”, “Monthly”, “Weekly”, “Daily” and “Hourly” format in a manner consistent for data Analysis or Charting and/or aligning with other market data. See FIGS. 4A-4D (discussed further below). In general, an objective of weather alpha may include providing one or more tools to quickly understand the vast amount of streaming data specifically related to changing weather perceptions and a magnitude of weather forecast changes directly to (changing) prices of any traded instruments (e.g., on an electronic exchange system and/or offline of an electronic exchange system).

[0087] The example three primary weather perspective workflows may include: [0088] 1)

Guidance: A system which allows for a data display, or charting display, of a weather forecast variable (like temperature, Degree Day, GWDD, wind, humidity, etc. . . .) from MULTIPLE weather forecast models, initialized and released at one or more specific times of day (model runs) from which a specific location or locations (e.g., city or region) which may (or may not) include preliminary estimated forecast values such as ‘% complete’ or ‘Fast& Full’ or partial week, month or seasonal aggregations called “Progress’ or pre-calculated (streaming) value(s) (e.g., “Change” from previous and/or “Anomaly/Difference” from Climatology) to be displayed in real-time or its history, allowing the user to closely examine weather forecast risks, by visualizing if all weather forecasts are in agreement or disagreement with each other over the forecast duration. The ‘weather Symbology’ of the present disclosure allows the user a method to easily access weather forecast data in a manner consistent for data Analysis or Charting and/or aligning with other market data.

See FIG. 5 (discussed further below). In some examples, Daily or Hourly represent weather forecast fields. The symbology allows weather forecasts to be treated as a continuation contract (e.g., such as a prompt month natural gas contract) or revert to previous weather forecasts. Data (including in some examples all data) in DID system **102** may be displayed as a function of receipt time or interval start time. In some examples, the Symbology may include an MRO format (described herein). [0089] 2) Progression: This style of chart may hold a particular Forecast Valid Date/Time of Day on that specific Date constant (called a “Target Date”) and shows how the weather forecast models have changed overtime, leading up to that “target date”. The system which allows for either a data display or charting display of a weather forecast variable (like temperature, Degree Day, GWDD, wind, humidity, etc.) from MULTIPLE weather forecast models, and any number of model runs, for a specific location (city or region) which may (or may not) include preliminary estimated values such as ‘% complete’ or ‘Fast& Full’ or partial week, month or seasonal aggregations called “Progress’ or pre-calculated forecast value (i.e. “Change” from previous and/or “Anomaly/Difference” from Climatology) to be displayed in real-time, allowing the user to closely examine weather forecast changes over time (to a point at which they all may converge). The ‘weather Symbology’ of the present disclosure allows the user a method to easily access weather forecast data in either a “Seasonal”, “Monthly” “Weekly”, “Daily” or “Hourly” format in a manner consistent for data Analysis and/or Charting and aligning with other market data. See FIGS. 6A-6C (discussed further below). [0090] 3) Accuracy: This style of chart or data query may look back at historical weather forecast ‘model run dates’ and ‘model runs’ and hold a Forecast Valid Day constant. For example ‘Forecast Day #1’ could be examined over the last 30 days of weather forecasts from a particular weather forecast model. Each of these Day 1 weather forecasts (which refer to a different Forecast Valid Date) can then be subtracted from their respective weather observation (then all (e.g., 30) data points may be averaged) to derive an average weather forecast Bias (hotter/colder) or other performance statistics. The ‘weather Symbology’ of the present disclosure allows the user a method to easily access weather forecast data in either a “Daily” or “Hourly” format in a manner consistent for data Analysis or Charting and/or aligning with other market data. See FIGS. 7A and 7B (discussed further below).

Overview of Weather Symbology

[0091] To accomplish the example weather perspective and weather alpha workflows (which may be important, for example, to traders, analysts, meteorologist and trading floor quants, investors, hedgers), lies an underlying comprehensive ‘weather symbology’. This symbology structure is desirable for weather data to function inside a streaming time series server-based trading desktop (such as time series server **118**), where the time series server also accommodates a fast streaming delivery of market prices. It is not the order of the elements listed in the ‘weather symbology’ but rather the function of what weather ‘elements’ can be extracted and applied to client application(s)/service(s) **126** (such as a trading desktop) which hosts market prices. To describe the methodology of the ‘weather symbology’, a unique structure is established by the present disclosure to mimic each of the three primary weather perspective workflows. [0092] 1) Guidance based weather symbology may use a ‘location’, a ‘Weather Model Name’ along with a specific ‘Weather Model Run Date’ and a ‘Weather Model Run’ to extract any weather forecast, initialized and produced any time of day, for either current running forecasts or historical weather forecasts. Along with that functionality, users may have shortcut codes, meaning a user may obtain the most current weather forecast for that ‘Location’, ‘Weather Model Name’ or ‘Weather Model Run’. In addition, the Guidance based weather symbology structures may allow for either “specific” or “rolling” symbols, meaning a user can request a specific ‘Weather Model Run Date’ and/or specific ‘Model Run’ or “rolling”. From that core information, the Guidance weather symbol can be further extended to include calendar, seasonal, monthly, weekly hourly or daily weather forecast data fields, for any specific location, for any weather forecast variable, for any preliminary estimated values such as ‘% complete’ or ‘Fast& Full’ or progress, or pre-calculated values (e.g., “Change”

from previous and/or “Anomaly/Difference” from Climatology) or Ensemble based statistics in the symbol itself. All of this weather information can then be easily combined in a manner consistent for data Analysis, Charting and/or aligning with other market data in either “Exchange Time” or “Interval Start” time. See FIG. 5 and the description herein for constructing the weather symbology. [0093] 2) Progression based weather symbology may use a ‘location’, a ‘Weather Model Name’ and an optional ‘Weather Model Run’ to extract all weather forecasts initialized and produced for a specific “target Date”. The Progression based weather symbols can accommodate any weather forecast, produced any time of day, for current running forecasts and historical weather forecasts. Along with the functionality users may have shortcut codes, meaning a user may obtain the most current weather forecast for that ‘Location’, ‘Weather Model Name’ or ‘Weather Model Run’. In addition, the Progression based weather symbology structure may allow for either “specific” or ‘rolling symbols’, meaning a user can request a specific or rolling ‘Target Dates’. From that core information, the Progression weather symbol can be further extended to include calendar, seasonal, monthly, weekly daily or hourly weather forecast data, for any specific location, for any weather forecast variable, for any preliminary estimated values such as ‘% complete’ or ‘Fast& Full’ or Progress or pre-calculated values (e.g., “Change” from previous and/or “Anomaly/Difference” from Climatology) or Ensemble based statistics in the symbol itself. All of this weather information can then be easily combined in a manner consistent for data Analysis, Charting and/or aligning with other market data in either “Exchange Time” or “Interval Start” time. Sec FIGS. 6A-6C. [0094] 3) Accuracy based weather symbology may use a ‘location’, a ‘Weather Model Name’ and an optional ‘Weather Model Run’ to extract all weather forecasts produced for a specific time in the future “Forecast Valid Day or Model Hour”. This symbol may use the “!” for a rolling notation, so the user can always extract a particular ‘Forecast Valid Day’ in the future. For example a ‘Day 1’ may be a different “Date” with each new ‘Weather Model Run Date’, so Day 1 is rolling. The Accuracy based weather symbols can accommodate any weather forecast, produced any time of day, and operate during current running forecasts and historical weather forecasts. In addition the Accuracy based weather symbology structure may allow for ‘rolling symbols’ related to a “Forecast Valid Day” (e.g., as opposed to a specific ‘Forecast Valid Date’/like used in a Progression’s ‘Target Date’). From that core information, returned from the weather symbology, the Accuracy weather symbol can be further extended to include, for example, calendar, seasonal, monthly, weekly daily or hourly weather forecast data, for any specific location, for any weather forecast variable, for any ‘Fast& Full’ or Progress, or Ensemble based statistics in the symbol itself. All of this weather information can then be easily combined in a manner consistent for data Analysis, Charting and/or aligning with other market data in either “Exchange Time” or “Interval Start” time. See FIGS. 7A and 7B.

[0095] There are many sources of quality weather forecasts from various government entities, such as weather agencies, educational institutions, national research laboratories, military as well as private companies. The initial focus is from two primary government weather forecast entities; being the American government NOAA and European government ECMWF. Each of the two entities produces multiple weather forecast products (a few examples of these weather forecast models include GFS, GFS ENS, GFS ENS EXT, CFS ENS, ECMWF, ECMWF ENS, and ECMWF ENS EXT, SEAS, SEAS EXT), which generally have the greatest influence on market prices related to numerous traded instruments around the globe. Four of these weather forecast models are from the United States (GFS, GFS ENS, GFS ENS EXT, CFS ENS) and the remaining five (ECMWF, ECMWF ENS, ECMWF EXT, SEAS and SEAS EXT) are from Europe. Four of these weather Forecast Models are produced four times per day (Model Runs), the remaining five are produced in less frequent updates (e.g., daily or twice per week, etc. . . .).

[0096] Tables 1 and 2 below include a list of weather forecast models and weather forecast variables which are included in the ‘weather symbology’ of the present disclosure. In particular, Table 1 illustrates an example list of weather forecast models for the weather symbology of the

present disclosure. Table 2 illustrates an example list of weather forecast variables for the weather symbology of the present disclosure. In Table 2, H represents an Hourly variable and D represents a Daily variable.

[0097] The ‘weather symbology’ described in the present disclosure may be configured to operate with all nine of these example weather models, as well as any and all other weather forecasts, either in the public or private domain.

TABLE-US-00001 TABLE 1 Example list of weather forecast models for the weather symbology

CITY FORECAST DATA	Weather	Spatial	Forecast	Temporal	Model	Name	Code	Resolution
Domain	Model	Runs	Duration	Resolution	GFS	GFS	¼ degree	Global
120 (aka 18 mile/	Forecast	18z	0-384 hrs	3 hr	121-384	28 km	grid)	GFS
6z, 12z, 16 days	3 hr-192	Ensemble	(aka 18 mile/	Forecast	18z	0-384 hrs	6 hr	198-384
28 km	grid)	ECMWF	ECM	¼ degree	Global	0z, 12z	10 days	Hourly-90
(aka 18 mile/	Forecast	0-240 hrs	3 hr	93-144	28 km	grid)	6 hr	150-240
ECMWF	ECM	¼ degree	Global	6z, 18z	10 days	Hourly-90	(aka	18 mile/
Forecast	0-240 hrs	28 km	grid)	ECMWF	ECE	¼ degree	Global	6z, 18z
15 days	Hourly-90	ENS	(aka 18 mile/	Forecast	0-360 hrs	3 hr	93-144	28 km
grid)	6 hr	150-360	ECMWF	ECMR	¼ degree	Global	2× per week	16-46 days
Daily	ENS	EXT	(aka 18 mile/	Forecast	Monday &	28 km	grid)	Thursday
CFS	CFS	½ degree	Global	0z only	0-9 months	Daily	Forecast	SEAS
SEAS	SEAS	¼ degree	Global	1× month	0-7 months	Daily	(aka 18 mile/	Forecast
28 km	grid)	TABLE-US-00002	TABLE 2	Example (non-limiting)	list of a weather	forecast	variables	for the
weather	symbology.	Time	Frame	Field	Name	Front	End	Long
Short	WEATHER	META	DATA	SPECIFICS	Time	Receipt	Time	Receipt
Time	nstime	Nstime	nstime	ns_exchange	time	ns_exchange	time	ns_exchange
time	ns_exchange	time	Line_id	Line_id	Line_id	sequence_number	sequence_number	sequence_number
sequence_number	sequence_series	sequence_series	sequence_series	sequence_series	flags	Flags	flags	utcdatetime
UTC	Forecast	Valid	Date	and	UTC	Date	and	Time
Forecast	Valid	Tme	localdatetime	LOCAL	Forecast	Valid	Date	and
Local	Date	and	Time	Forecast	Valid	Tme	fastvfull	Ensemble
Member	fast	vs	full	ENS	Fast	or	Full	filetime
File	Time	File	Time	location	Location	Location	Model	Forecast
Model	Name	Model	Name	modeldate	Model	Run	Date	MR
Date	modelrun	Forecast	Model	Run	MR	forecasthour	Forecast	Model
Hour	Mhr	interpolated	Interpolated	Value	Intrpl	blended	Forecast	and
METAR	are	Blended	Blended	localdate	Local	Date	Local	Date
localhour	Local	Hour	Local	Hr	ensemblevariable	Ensemble	Variable	Ens
Varbl	utcdte	UTC	Date	UTC	Date	utchour	UTC	Hour
UTC	Hr	localforecastday	Local	Forecast	Day	Local	Day	percentcomplete
Percent	Complete	%	Comp	LIQUID	PRECIPITATION	H	APCP	Accum
Precip	6 hr	reset	APCP	H	APCPNO	Accum	Precip	hourly
AccPrecip	H	Intrpl	Interpolated	D	DAYPRECIP	Daily	Precipitation	Daily
Precip	D	DAYPRECIP_Chng_6	h	Daily	Precip	6 Hr	Change	D
Precip	6 hr	CHG	D	DAYPRECIP_Chng_12	h	Daily	Precip	12 Hr
Change	D	Precip	12 hr	CHG	D	DAYPRECIP_Chng_18	h	Daily
Precip	18 Hr	Change	D	Precip	18 hr	CHG	D	DAYPRECIP_Chng_24
h	Daily	Precip	24 Hr	Change	D	Precip	24 hr	CHG
D	DAYPRECIP_Diff_10Y	Avg	Daily	Precip	10 yr	Anomaly	D	Precip
10 yr	ANOM	D	DAYPRECIP_Diff_15Y	Avg	Daily	Precip	15 yr	Anomaly
15 yr	ANOM	D	DAYPRECIP_Diff_30Y	Avg	Daily	Precip	30 yr	Anomaly
30 yr	ANOM	D	DAYPRECIP_Diff_30Y	Climo	Dally	Precip	Official	30 yr
D	Precip	o30 yr	ANOM	Anomaly	SNOW & RUNOFF	H	WATR	Snow
Water	Runoff	SWRO	H	WATRNO	Snow	Water	Runoff	Hourly
Runoff	H	Intrpl	interpolated	D	DAYWATR	Daily	Snow	Water
Runoff	D	Runoff	D	WEASD	Snow	Water	Equivalent	SWE
H	WEASDNO	Snow	Water	Equivalent	Hourly	SWE	H	Intrpl
Interpolated	D	DAYWEASD	Daily	Snow	Water	Equivalent	D	SWE
H	SNOD	Snow	Depth	SD	D	SNOD_Avg	Daily	Avg
Snow	Depth	Avg	SD	SOLAR	H	DLWRF	Downward	Long
Wave	Radition	Long	Wave	Radition	LWRad	H	intrpl	Flux
Hourly	interpolated	D	DLWRF_Avg	Downward	Long	Wave	Radition	LWRad
D	Flux	Daily	Avg	H	DSWRF	Downward	Short	Wave
Radition	SWRad	H	Intrpl	Flux	Hourly	Interpolated	D	DSWRF_Avg
Downward	Short	Wave						

Radiation SWRad D Flux Daily Avg H GFLUX Surface Ground Heat Flux Grnd Heat Flux H Intrpl
W/m{circumflex over ()}2 H GFLUXNO Surface Ground Heat Flux Grnd Heat Flux H Intrpl
W/m{circumflex over ()}2 Hourly Intrpl H WILT Wilt hourly H Wilt D WILT_Avg Daily Avg
Surface Ground Heat D Avg Gmd Heat Flux Flux W/m{circumflex over ()}2 WIND SPEED &
DIRECTION H GUST Wind Gust max per hour Gust hr H WINDS_10 m 10 Meter Wind Speed
10M W Speed H WINDD_10 m 10 Meter Wind Direction 10M W Dir H TMP_80 m 80M Temp 80
m T H PRES_80 m 80 m Pressure 80 m Pres H SPFH_80 m 80M Specific Humidity 80 m SpcH H
WINDS_80 m 80M Wind Speed 80 m WS H WINDD_80 m 89M Wind Direction 80 m WD
AGRICULTURE D GUST_D_MAX Daily Avg Wilt D avg Wilt H SOILW_0_0.1 m Soil Moisture
0 cm 100 cm deep Soil Mo 0-100 cm D SOILW_0_0.1 m_Avg Daily Avg Soil Moisture 0 cm-
DAvg Soil Mo 0-100 cm 100 cm deep H SOILW_0.1_0.4 m Soil Moisture 100 cm-400 cm Soil Mo
100 cm-400 cm deep D SOILW_0.1_0.4 m_Avg Daily Avg Soil Moisture 100 cm- D Avg Soil Mo
100 cm- 400 cm deep 400 cm H SOILW_0 4_1 m Soil Moisture 400 cm-1 m deep Soil Mo 400
cm-1 m D SOILW_0.4_1 m_Avg Daily Avg Soil Moisture 400 cm- D Avg Soil Mo 400 cm-1 m 1 m
deep H SOILW_1_2 m Soil Moisture 1 m-2 m deep Soil Mo 1 m-2 m D SOILW_1_2 m_Avg Daily
Avg Soil Moisture 1 m-2 m D Avg Sod Mo 1 m-2 m deep H TSOIL_0_0.1 m Soil Temp 0 cm-100
cm deep Soil T 0-100 cm D TSOIL_0_0.1 m_Avg Daily Avg Soil Temp 0 cm- D Avg Soil T 0-100
cm 100 cm deep H TSOIL_0.1_0.4 m Soil Temp 100 cm-400 cm deep Soil T 100 cm-400 cm D
TSOIL_0.1_0.4 m_Avg Daily Avg Soil Temp 100 cm- D Avg Soil T 100 cm-400 cm 400 cm deep H
TSOIL_0.4_1 m Soil Temp 400 cm-1 m deep Soil T 400 cm-1 m D TSOIL_0.4_1 m_Avg Daily Avg
Soil Temp 400 cm-1 m D Avg Soil T 400 cm-1 m deep H TSOIL_1_2 m Soil Temp 1 m-2 m deep
Soil T 1 m-2 m D TSOIL_1_2 m_Avg Daily Avg Soil Temp 1 m-2 m D Avg Soil T 1 m-2 m deep
SKY & SURFACE - HOURLY H TCDC_ALL Total Cloud Cover entire Cloud All atmosphere H
TCDC_HIGH Total Cloud Cover highest level Cloud High H TCDC_LOW Total Cloud Cover
lowest level Cloud Low H TCDC_MID Total Cloud Cover mid level Cloud Mid H PRMSL Mean
Sea Level Pressure MSLP H VIS Visibility Vis H DPT_2 m 2M Dew Point 2 m DP H RH_2 m 2M
Relative Humidity 2 m RH H SPFH_2 m Specific Humidity Spec Hum H TMP_Hourly_2 m 2M
Hourly Temp 2 m T Hourly H TMP_Hourly_2 m_Chng_12 h 2M Hourly Temp 12 Hr Change 2 m
hr T 12 hr CHG H TMP_Hourly_2 m_Chng_18 h 2M Hourly Temp 18 Hr Change 2 m hr T 18 hr
CHG H TMP_Hourly_2 m_Chng_24 h 2M Hourly Temp 24 Hr Change 2 m hr T 24 hr CHG H
TMP_Hourly_2 m_Chng_6 h 2M Hourly Temp 6 Hr Change 2 m hr T 6 hr CHG SKY &
SURFACE DAILY D TMP_D_Avt_2 m 2M Daily Avg Temp 2 m D Avg T D TMP_D_Avg_2
m_Chng_6 h 2M Daily Avg Temp 6 Hr 2 m D Avg T 6 hr CHG Change D TMP_D_Avg_2
m_Chng_12 h 2M Daily Avg Temp 12 Hr 2 m D Avg T 12 hr CHG Change D TMP_D_Avg_2
m_Chng_18 h 2M Daily Avg Temp 18 Hr 2 m D Avg T 18 hr CHG Change D TMP_D_Avg_2
m_Chng_24 h 2M Daily Avg Temp 24 Hr 2 m D Avg T 24 hr CHG Change D TMP_D_Avg_2
m_Diff_10Y Avg 2M Daily Avg Temp 10 yr 2 m D Avg T 10 yr ANOM Anomaly D TMP_D_Avg_2
m_Diff_15Y Avg 2M Daily Avg Temp 15 yr 2 m D Avg T 15 yr ANOM Anomaly D TMP_D_Avg_2
m_Diff_30Y Avg 2M Daily Avg Temp 30 yr 2 m D Avg T 30 yr ANOM Anomaly D TMP_D_Avg_2
m_Diff_o30Y Climo 2M Daily Avg Temp 2 m D Avg T o30 yr ANOM Official 30 yr Anomaly D
TMP_D_MAX_2 m 2M Daily Max Temp 2 m D_XT D TMP_D_MAX_2 m_Chng_6 h 2M Daily
Max Temp 6 Hr 2 m D_XT 6 hr CHG Change D TMP_D_MAX_2 m_Chng_12 h 2M Daily Max
Temp 12 Hr 2 m D_XT 12 hr CHG Change D TMP_D_MAX_2 m_Chng_18 h 2M Daily Max
Temp 18 Hr 2 m D_XT 18 hr CHG Change D TMP_D_MAX_2 m_Chng_24 h 2M Daily Max
Temp 24 Hr 2 m D_XT 24 hr CHG Change D TMP_D_MAX_2 m_Diff_10Y Avg 2M Daily Max
Temp 10 yr 2 m D_XT 10 yr ANOM Anomaly D TMP_D_MAX_2 m_Diff_15Y Avg 2M Daily Max
Temp 15 yr 2 m D_XT 15 yr ANOM Anomaly D TMP_D_MAX_2 m_Diff_30Y Avg 2M Daily Max
Temp 30 yr 2 m D_XT 30 yr ANOM Anomaly D TMP_D_MAX_2 m_Diff_30Y Climo 2M Daily
Max Temp 2 m D_XT o30 yr ANOM Official 30 yr Anomaly D TMP_D_Min_2 m 2M Daily Min
Temp 2 m D_NT D TMP_D_Min_2 m_Chng_12 h 2M Daily Min Temp 12 Hr 2 m D_NT 12

hrCHG Change D TMP_D_Min_2 m_Chng_18 h 2M Daily Min Temp 18 Hr 2 mD_NT 18 hr CHG
Change D TMP_D_Min_2 m_Chng_24 h 2M Daily Min Temp 24 Hr 2 mD_NT 24 hrCHG Change
D TMP_D_Min_2 m_Chng_6 h 2M Daily Min Temp 6 Hr 2 mD_NT 6 hr CHG Change D
TMP_D_Min_2 m_Diff_10YAvg 2M Daily Min Temp 10 yr 2 mD_NT 10 yr ANOM Anomaly D
TMP_D_Min_2 m_Diff_15YAvg 2M Daily Min Temp 15 yr 2 mD_NT 15 yr ANOM Anomaly D
TMP_D_Min_2 m_Diff_30YAvg 2M Daily Min Temp 30 yr 2 mD_NT 30 yr ANOM homely D
TMP_D_Min_2 m_Diff_30YCIimo 2M Daily Min Temp 2 mD_NT o30 yANOM Official30 yr
Anomaly DEGREE DAYS - DAILY D HDD Heating Degree Day 65 degrees HDD D CDD
Cooling Degree Day 65 degrees CDD Gas Weeks Heating Degree Day 65 degrees HDD NGwk
HDD_NG Wks (NGwk) Gas Weeks Cooling Degree Day 65 degrees CDD NGwk CDD NG Wks
(NGwh| D GWDD GWDD GWDD D GWDD_Chng_6 h GWDD 6 Hr Change GWDD 6 hr CHG
D GWDD_Chng_12 h GWDD 12 Hr Change GWDD 12 hr CHG D GWDD_Chng_18 h GWDD
18 Hr Change GWDD 18 hr CHG D GWDD_Chng_24 h GWDD 24 Hr Change GWDD 24 hr
CHG D GWDD_Diff_10YAvg GWDD 10 yr Anomaly GWDD 10 yr ANOM D
GWDD_Diff_15YAvg GWDD 15 yr Anomaly GWDD 15 yr ANOM D GWDD_Diff_30YAvg
GWDD 30 yr Anomaly GWDD 30 yr ANOM D GWDD_Diff_o30YAvgDIFF GWDD Official 30
yr Anomaly GWDD o30 yrANOM Gas Weeks Gas Weighted Temp (NG weeks) GW Temp
(NGwk) GWDD_NG Wks Gas Weeks Gas Weighted Temp 12 Hr GW T 12 hr CHG (NGwk)
GWDD_Chng 6 h_NG Wks CHNG (NGwk) Gas Weeks Gas Weighted Temp 18 Hr GW T 18 hr
CHG (NGwk) GWDD_Chng 12 h_NG Wks CHNG (NGwk) Gas Weeks Gas Weighted Temp 24
Hr GW T 24 hr CHG (NGwk) GWDD_Chng 18 h_NG Wks CHNG (NGwk) Gas Weeks Gas
Weighted Temp 6 Hr CHNG GW T 6 hr CHG (NGwk) GWDD_Chng 24 h_NG Wks (NGwk) Gas
Weeks Gas Weighted Temp 10 yr GW T 10 yr ANOM (NGwk) GWDD_Diff_10 YAvg_NG Wks
Anomaly (NGwk) Gas Weeks Gas Weighted Temp 15 yr GW T 15 yr ANOM (NGwk)
GWDD_Diff_15 YAvg_NG Wks Anomaly (NGwk) Gas Weeks Gas Weighted Temp 30 yr GW T
30 yr ANOM (NGwk) GWDD_Diff_30 YAvg_NG Wks Anomaly (NGwk) Gas Weeks Gas
Weighted Temp Official GW T o30 yr ANOM GWDD_Diff_o30 YAvgDIFF_NG Wks 30 yr
Anomaly (NGwk) (NGwk) D GWAvgT Gas Weighted Daily Avg Temp GW Avg Temp D
GWAvgT_Chng_12 h Gas Weighted Avg Tamp 12 Hr GW AvgT 12 hr CHG CHNG D
GWAvgT_Chng_18 h Gas Weighted Avg Tamp 18 Hr GW AvgT 18 hr CHG CHNG D
GWAvgT_Chng_24 h Gas Weighted Avg Temp 24 Hr GW AvgT 24 hr CHG CHNG D
GWAvgT_Chng_6 h Gas Weighted Avg Temp 6 Hr GW AvgT 6 hr CHG CHNG D
GWAvgT_Diff_10YAvg Gas Weighted Avg Temp 10 yr GW AvgT 10 yr ANOM Anomaly D
GWAvgT_Diff_15YAvg Gas Weighted Avg Temp 15 yr GW AvgT 15 yr ANOM Anomaly D
GWAvgT_Diff_30YAvg Gas Weighted Avg Tamp 30 yr GW AvgT 30 yr ANOM Anomaly D
GWAvgT_Diff_o30VAvgDIFF Gas Weighted Avg Temp GW AvgT o30 yr ANOM Official 30 yr
Anomaly Gas Weeks Gas Weighted Avg Temp (NG GW Avg Temp (NGwk) GWAvgT_NG Wks
weeks) Gas Weeks Gas Weighted Avg Temp 12 Hr GW AvgT 12 hr CHG GWAvgT_Chng_12
h_NC Wks CHNG (NGwk) (NGwk) Gas Weeks Gas Weighted Avg Temp 18 Hr GW AvgT 18 hr
CHG GWAvgT_Chng_18 h_NG Wks CHNG (NGwkl (NGwk) Gas Weeks Gas Weighted Avg
Temp 24 Hr GW AvgT 24 hr CHG GWAvgT.Chng_24 h.NG Wks CHNG (NGwk| (NGwk) GM
Weeks GWAvgT Gas Weighted Avg Temp 6 Hr GW AvgT 6 hr CHG Chng_6 h NG Wks CHNG
(NGwk) (NGwk) Gas Weeks Gas Weighted Avg Tamp 10 yr GW AvgT 10 yr ANOM
GWAvgT_Diff_10 YAvg_NG Wks Anomaly (NGwb| (NGwk) Gas Weeks Gas Weighted Avg
Temp 15 yr GW AvgT 15 yr ANOM GWAvgT_Diff_15 YAvg_NG WKt Anomaly (NGwk) (NGwk)
Gas Weeks Gas Weighted Avg Temp 30 yr GW AvgT 30 yr ANOM GWAvgT_Diff_30 YAvg_NG
WKt Anomaly (NGwk) INGwk) Gas Weeks Gas Weighted AvgTemp Official GW AvgT o30 yr
ANOM GWAvgT_Diff_o30 YAvgDIFF_NG Wks 30 yr Anomaly (NGwk) (NGwk) D GWMaxT
Gas Weighted Max Temp GW Max Temp D GWMaxT_Chng_12 h Gas Weighted Max Temp 12 Hr
GW XT 12 hr CHG CHNG D GWMaxT_Chng_18 h Gas Weighted Max Temp 18 Hr GW XT 18

hr CHG CHNG D GWMaxT_Chng_24 h Gas Weighted Max Temp 24 Hr GW XT 24 tM CHG
CHNG D GWMaxT_Chng_6 h Gas Weighted Max Temp 6 Hr GW XT 6 hr CHG CHNG D
GWMaxT_Diff_10YAvg Gas Weighted Max Temp 10 yr GW XT 10 yr ANOM Anomaly D
GWMaxT_Diff_15YAvg Gas Weighted Max Temp 15 yr GW XT 15 yr ANOM Anomaly D
GWMaxT_Diff_30YAvg Gas Weighted Max Temp 30 yr GW XT 30 yr ANOM Anomaly D
GWMaxT_Diff_o30YAvgDIFF Gas Weighted Max Temp GW XT o30 yr ANOM Official 30 yr
Anomaly Gas Weeks Gas Weighted Max Temp (NG GW Max Temp (NGwk) GWMaxT_NG WKs
weeks) Gas Weeks Gas Weighted Max Temp 12 Hr GW XT 12 hr CHG (NGwk)
GWMaxT_Chng_12 h.NG WKs CHNG (NGwk) Gas Weeks Gas Weighted Max Temp 18 Hr GW
XT 18 hr CHG (NGwk) GWMaxT_Chng18 h.NG WKs CHNG (NGwk) Gas Weeks Gas Weighted
Max Temp 24 Hr GW XT 24 hr CHG (NGwk) GWMaxT_Chng_24 h.NG WKt CHNG (N6wk)
Gas Weeks Gas Weighted Max Temp 6 Hr GW XT 6 hr CHG (NGwk) GWMaxT_Chng_6 h.NG
WKs CHNG (NGwk) Gas Weeks Gas Weighted Max Temp 10 yr GW XT 10 yr ANOM
GWMaxT_Diff_10 YAvg_NG WKs Anomaly INGwk) (NGwk) Gas Weeks Gas Weighted Max
Temp 15 yr GW XT 15 yr ANOM GWMaxT_Diff_15 YAvg_NG WKs Anomaly (NGwk) INGwk)
Gas Weeks Gas Weighted Max Temp 30 yr GW XT 30 yr ANOM GWMaxT_Diff_30 YAvg_NG
WKs Anomaly (NGwk) (NGwk) Gas Weeks Gas Weighted Max Temp GW XT o30 yr ANOM
GWMaxT_Diff_o30 YAvgCLIMO_NG WKs Official 30 yr Anomaly (NGwk) (NGwk) D
GWMinT Gas Weighted Min Temp GW Min Temp D GWMinT_Chng_12 h Gas Weighted Min
Temp 12 Hr GW NT 12 hr CHG CHNG D GWMinT_Chng_18 h Gas Weighted Min Temp 18 Hr
GW NT 18 hr CHG CHNG D GWMinT_Chng_24 h Gas Weighted Min Temp 24 Hr GW NT 24 hr
CHG CHNG D GWMinT_Chng_6 h Gas Weighted Min Temp 6 Hr GW NT 6 hr CHG CHNG D
GWMinT_Diff_10YAvg Gas Weighted Min Temp 10 yr GW NT 10 yr ANOM Anomaly D
GWMinT_Diff_15YAvg Gas Weighted Min Temp 15 yr GW NT 15 yr ANOM Anomaly D
GWMinT_Diff_30YAvg Gas Weighted Min Temp 30 yr GW NT 30 yr ANOM Anomaly D
GWMinT_Diff_o30YAvgDIFF Gas Weighted Min Temp GW NT o30 yr ANOM Official 30 yr
Anomaly Gas Weeks Gas Weighted Min Temp (NG GW Min Temp (NGwk) GWMinT_NG WKs
weeks) Gas Weeks Gas Weighted Min Temp 12 Hr GW NT 12 hr CHG (NGwk)
GWMinT_Chng_12 h_NG WKs CHNG (NGwk) Gas Weeks Gas Weighted Min Temp 18 Hr GW
NT 18 hr CHG (NGwk) GWMinT_Chng_18 h_NG WKs CHNG (NGwk) Gas Weeks Gas
Weighted Min Temp 24 Hr GW NT 24 hr CHG (NGwk) GWMinT_Chng_24 h_NG WKs CHNG
(NGwk) Gas Weeks Gas Weighted Min Temp 6 Hr GW NT 6 hr CHG (NGwk) GWMinT_Chng_6
h_NG WKs CHNG (NGwk) Gas Weeks Gas Weighted Min Temp 10 yr GW NT 10 yr ANOM
GWMinT_Diff_10 YAvg_NG WKs Anomaly (NGwk) (NGwk) Gas Weeks Gas Weighted Min
Temp 15 yr GW NT 15 yr ANOM GWMinT_Diff_15 YAvg_NG WKs Anomaly (NGwk) (NGwk)
Gas Weeks Gas Weighted Min Temp 30 yr GW NT 30 yr ANOM GWMinT_Diff_30 YAvg_NG
WKs Anomaly (NGwk) (NGwk) Gas Weeks Gas Weighted Min Temp GW NT o30 yr ANOM
GWMinT_o30AvgDIFF_NG WKs Official 30 yr Anomaly (NGwk) [NGwk]

[0098] Components of DID system **102** (FIG. 1) and/or weather integration server **116** (FIG. 2) of the present disclosure may be embodied on a single computing device. In other examples, components of DID system **102** and/or weather integration server **116** may be embodied on two or more computing devices distributed over several physical locations, connected by one or more wired and/or wireless links. It should be understood that DID system **102** of the present disclosure refers to a computing system having sufficient processing and memory capabilities to perform the specialized functions described herein.

[0099] Some portions of the present disclosure describe embodiments in terms of algorithms and/or routines and symbolic representations of operations on information. These algorithmic descriptions and representations are used to convey the substance of this disclosure effectively to others skilled in the art. These operations, while described functionally, computationally, or logically, are to be understood as being implemented by data structures, computer programs or equivalent electrical

circuits, microcode, or the like. Furthermore, at times, it may be convenient to refer to these arrangements of operations as routines or algorithms. The described operations and their routines/algorithms may be embodied in specialized software, firmware, specially-configured hardware or any combinations thereof.

[0100] Those skilled in the art will appreciate that DID system **102** (of FIG. **1**) may be configured with more or less components to conduct the methods described herein with reference to FIGS. **3A** and **3B**. The methods described herein (that may be conducted by DID system **102** of the present disclosure) may be performed by processing logic that may comprise hardware (e.g., circuitry, dedicated logic, programmable logic, microcode, etc.), software (such as instructions run on a processing device), or a combination thereof. In one embodiment, the methods described herein may be performed by one or more specialized processing components associated with components **204-216** of weather integration server **116** of FIG. **2**. With respect to FIGS. **3A** and **3B**, although these flowcharts may illustrate a specific order of method steps, it is understood that the illustrated methods are exemplary, and that the order of these steps may differ. Also, in some examples, two or more steps may be performed concurrently or with partial concurrence.

[0101] FIG. **3A** is a flowchart diagram of an example method **300** of generating integrated weather and market data associated with data distribution environment **100** of FIG. **1**, and weather integration server **116** and time series server **118** of FIG. **2**, according to an aspect of the present disclosure.

[0102] At step **302**, weather integration server **116** may store a pre-defined weather symbology (e.g., in storage **214**). The weather symbology may include one or more pre-defined symbol elements, one or more variables and one or more pre-defined rules for combining and/or arranging elements among the symbol elements to form a weather symbology instruction having a pre-defined instruction structure. As discussed herein, the inclusion of one or more symbol elements in the pre-defined instruction structure may be used to create a variety of weather symbology instructions. In general, a weather symbology instruction may include symbol elements including at least one weather observation location, a weather structure (such as a weather perspective) and weather forecast model information. The weather symbology instruction may include additional variable information (such as condition(s) and/or function(s)) for generating desired weather forecast/observations data that may be integrated with market data, and that may provide actionable knowledge. In some examples, the symbol elements may be linked to different segments of weather data stream(s) associated with weather data source(s) **104**.

[0103] At step **304**, data collector **112** may collect weather data among weather data source(s) **104**. In some examples, the weather data may include weather observations data for one or more observation locations and weather forecast model data for one or more weather forecast models (including, in some examples, different model runs). At step **306**, data collector **112** may collect market data among market data source(s) **106**. In some examples, collected weather data (step **304**) may be distributed to weather integration server **116** and collected market data (step **306**) may be distributed to time series server **118** (e.g., by data feed distributor **114**). In some examples, weather integration server **116** may segment the collected weather data (step **304**) into one or more segments based on the weather symbology (stored in step **302**). It is understood that both the weather data and the market data may change over time and that steps **304** and **306** may include repeatedly collecting the weather and market data respectively, as the corresponding data changes. It is also understood that updates to the weather data may occur at different frequencies than updates to the market data. In some examples, the collection of weather and market data (steps **304** and **306**) may include the capture of (real-time) weather data stream(s) and (real-time) market data stream(s) (e.g., via one or more data feeds).

[0104] At step **308**, DID system **102** may generate interactive GUI **120** including weather impact dashboard **122** for display on user device(s) **108**. In some examples, a configuration of weather impact dashboard **122** may depend on an underlying client application/service **126** for which it is

launched (e.g., a trading desktop, a mobile application, a spreadsheet application, an API, etc.). At step **310**, time series server **118** may receive a weather data request (e.g., data request **224**) from user device(s) **108** via interactive GUI **120**.

[0105] At step **312**, time series server **118** may determine a weather symbology instruction based on the received weather data request (step **310**). In some examples, the weather data request may directly include the weather symbology instruction. In some examples, time series server **118** may convert the weather data request into the weather symbology instruction having the pre-defined instruction structure, in accordance with the pre-defined rules and pre-defined symbology elements (step **302**) stored in storage **214**.

[0106] At step **314**, time series server **118** may determine a pre-defined weather structure (e.g., a perspective) from the weather symbology instruction (determined in step **312**). At step **316**, time series server **118** may create a weather forecast (and/or observations) dataset (also referred to as weather forecast/observations dataset) from among the collected weather data, based on the weather symbology instruction including the weather structure indicated in the weather symbology instruction. For example, time series server **118** may obtain the corresponding weather perspective among the weather structure(s) (from storage **216**), and may extract at least a portion of the segmented weather data **218** (from among the collected observation data and at least a portion of the weather forecast model data) based on the symbol elements/variable(s) in the weather symbology instruction. The extracted weather forecast/observation(s) data associated with the weather symbology instruction and any other data and/or information associated with the corresponding weather structure may then form the weather forecast/observations dataset. The weather forecast/observations dataset may also be formed as a function of exchange time and interval time, for ease of integration with the collected market data.

[0107] At step **318**, time series server **118** may generate an integrated presentation package comprising a combination of the weather forecast/observations dataset (step **316**) and a portion of the collected market data (step **306**), configured in such a manner so that the weather forecast/observations dataset is integrated with the portion of the collected market data. At step **320**, the integrated presentation package (step **318**) may be presented on interactive GUI **120** via weather impact dashboard **122**. In some examples, the integrated presentation package may be presented in such a manner to permit user interaction, including, in some examples, user-customization of the integrated presentation package. At optional step **322**, DID system **102** may expose the integrated presentation package to one or more APIs. In some examples, step **320** may not be performed and step **322** may be performed instead. In some examples, steps **308** may not be performed and step **310** may include receiving a weather data request via user device(s) **108**, without being received via an interactive GUI (e.g., via client application(s)/service(s) **126**).

[0108] At step **324**, the integrated presentation package may be updated (e.g., on weather impact dashboard **122**, via one or more APIs) concurrent with changes to the weather data, the market data and/or user input (including changes in real-time). In some examples, the weather and market data may each include real-time data streams, such that steps **318** and **320** (and/or optional step **322**) may generate and present the presentation package in real-time, using real-time streaming data.

[0109] FIG. **3B** is a flowchart diagram of an example method **330** of generating integrated weather and market data for streaming data associated with data distribution environment **100** of FIG. **1**, and weather integration server **116** and time series server **118** of FIG. **2**, according to an aspect of the present disclosure.

[0110] At step **332**, a pre-defined weather symbology may be stored, for example, in storage **214**. As described herein, the weather symbology may include pre-defined symbol elements that may be linked to one or more segments of one or more weather data streams. In some examples, storage **214** may also store one or more rules for generating weather symbology instruction(s). In some examples, DID system **102** may include a pre-defined market symbology, similar to the weather symbology, that may be linked to one or more segments of one or more market data streams.

[0111] At step **334**, one or more data feeds to weather data source(s) **104** and market data source(s) **106** may be initiated, for example, via data collector **112**. In example method **330**, weather data source(s) **104** and market data source(s) **106** may provide streaming data via corresponding data feed(s). In some examples, data collector **112** may also receive other data/information from among weather data source(s) **104** and market data source(s) **106** in a non-streaming manner, such as, periodically, in response to an event, condition and the like. At step **336**, data collector **112** may receive one or more weather data streams from weather data source(s) **104** (via respective data feed(s) initiated in step **334**). At step **338**, data collector **112** may receive one or more market data streams from market data source(s) **106** (via respective data feed(s) initiated in step **334**).

[0112] At step **340**, the received weather data stream(s) (in step **336**) may be segmented (e.g., separated and/or indexed), for example, by weather integration server **116**, based on the weather symbology (stored in step **332**), to form segmented (streaming) weather data. For example, the weather data stream(s) may be segmented in accordance with the (linked) pre-defined symbol elements. In this manner, a particular segment of the segmented weather data that is linked to a particular pre-defined symbol element may be extracted and the extracted (streaming) segment used to create an integrated presentation package, stream to user device(s) **108**, provided to an external system for further analysis and the like. In some examples, time series server **118** may be configured to segment the received market data stream(s) into one or more segments based on a pre-defined market symbology.

[0113] At step **342**, time series server **118** may receive a weather data request (e.g., via interactive GUI **120**, via client application(s)/service(s) **126**, etc.). At step **344**, time series server **118** may identify one or more requested symbol elements (and in some examples, one or more variables) in the received weather data request (step **344**). At step **346**, time series server **118** may create at least one weather symbology instruction based on the identified symbol element(s) (and, in some examples any variables) as identified in step **344**. In some examples, the received weather data request (step **342**) may directly include in a weather symbology instruction, so that steps **344** and **346** may be skipped.

[0114] At step **348**, time series server **118** may extract one or more segments of the segmented weather data stream(s) according to the weather symbology instruction (step **346**) to form one or more extracted weather stream(s) associated with the weather symbology instruction. In particular, the identified symbol elements, that are linked to corresponding segments of the segmented weather data stream(s) may be used to extract particular segment(s) of the segmented weather data stream(s). In some examples, any variables (e.g., conditions, functions) in the weather symbology instruction may be used to further extract segment(s) and/or portion(s) of segment(s) of the segmented weather data stream(s).

[0115] At step **350**, time series server **118** may create a real-time (streaming) weather dataset from the extracted weather stream(s), based on the weather data request (step **342**). More particularly, a weather structure that may be indicated in the weather symbology instruction (determined from the weather data request) may be obtained from storage **216**, and may be used to create the real-time weather dataset from the extracted weather stream(s). The real-time weather dataset may be formed as a function of exchange time and interval time, for ease of integration with the market data stream(s).

[0116] At step **352**, time series server **118** may generate an integrated presentation package of the real-time weather dataset and one or more market data stream(s) (received at step **338**). In some examples, time series server **118** may also extract one or more segments among similarly segmented market data stream(s) based on any market data symbology (e.g., indicated in the weather data request, based on pre-defined parameter(s) set by DID system **102**, based on pre-defined parameter(s) indicated by the user, and the like).

[0117] At step **354**. DID system **102** may expose the integrated presentation package to one or more APIs. At optional step **356**, the integrated presentation package may be presented on weather

impact dashboard of interactive GUI **120**. In some examples, step **354** may not be performed and step **356** may be performed instead. In some examples, both steps **354** and **356** may be performed. [0118] At step **358**, the integrated presentation package may be updated (e.g., via one or more APIs, on weather impact dashboard **122**, etc.) concurrent with changes to the weather data stream(s), the market data stream(s) and/or user input.

[0119] Referring to FIG. **3C**, hierarchical diagram **360** illustrating example weather symbology **362** and use of weather symbology **362** to form various presentation packages **370** are shown. In general, weather symbology **362** may include one or more weather symbol elements **364** (e.g., weather model sources, weather model locations and weather functions also referred to as weather perspectives). Weather symbology **362** may also include one or more weather variables **366** (e.g., functions and/or conditions). Symbol element(s) **364** and weather variable(s) **366** may be linked to particular segments of one or more weather data streams. A combination of symbol element(s) **364** and weather variable(s) **366** may be arranged into at least one weather symbology instruction (having the instruction structure described herein) in order to extract the particular linked segments of the weather data stream(s) to form a weather dataset. Similarly, time series server **118** may use market data symbology **368** (symbol elements/variables associated with, for example, commodity prices, supply/demand data, equity prices, earning per share data, fixed income data, municipal bonds, etc.) to extract particular segments from market data stream(s). Time series server **118** may combine the weather dataset (from among weather symbology **362** and market data symbology **368**) to form one or more presentation packages **370** (e.g., charts, watchlists, tables, grids, alerts, data feeds, data associated with object oriented programming languages (e.g., Python™), spreadsheet-formatted data (such as with RTD links), etc.). Presentation package(s) **370** may be presented to user device(s) **108** via client application(s) service(s) **126**. In some examples, presentation packages(s) **370** may be further configured into one or more weather workbooks **372** (e.g., natural gas (North America (NA), Europe (EU)), power, global crude oil, etc.) for presentation on trade desktop **374**.

[0120] In general, weather symbology **362** may be configured to instantly convert weather data into accurate and actionable knowledge, tailored specifically for traded instruments, via one or more (interactive) presentation packages. In some examples, weather data may be accessed in real-time using weather symbology **362**. Because of the weather symbology of the present disclosure, extremely large streams of weather and price data may be converted into Actionable Knowledge (such as presentation packages **370**, weather workbooks **372** and the like). Of significance, weather symbology **362** provides a mechanism to rapidly break down streaming weather data into the smallest possible building blocks in a manner configured to be integrated with market data handled by time series server **118**, and for interactive presentation on trading desktop **374**. The actionable knowledge (e.g., components **370**, **372**) created by DID system **102** allows users to instantly and easily interact with both weather and price data streams (e.g., including through time and space). Still further, weather symbology **362** allows user to create any user-customizable weather display. In some examples, weather symbology **362** may be configured to treat weather data similar to a traded instrument, so that weather symbol element **364**/weather variable **366** may be entered directly into a chart, table, watchlist, etc. . . . to view the weather data directly. Yet further, weather symbology **362** also allow the weather data to be accessed from multiple directions (e.g., three separate directions, catering to (in some examples) three primary weather workflows (e.g., guidance, progression and accuracy).

Description of Weather Symbology

[0121] Next, a detailed description of the weather symbology of the present disclosure is provided.

[0122] 1) General: All symbols (also referred to as symbol elements) have the same basic structure.

In general, a weather symbology instruction includes a same pre-defined weather symbology instruction structure: [0123] a. Location {space} Function {underscore}_ Model Run {Dash}-Wx

Model. {dot}field name [0124] i. EX: KORD MR0!_12z-GFS [0125] ii. wx variables can be

added at the end of the weather symbol [0126] iii. Model Run is an optional field [0127] iv. Dates are expressed as YYMMDD (2 digits each) [0128] 1. If a specific hour of the day (or model hour) is required, it accepts 1-3 digits: a. YYMMDDHHH [0129] v. Model runs (2 digits along with a “z”, for example 06z) [0130] vi. Rolling Symbols have a bang “!” [0131] 2) Guidance (aka Model Runs): All weather symbols below can access any time aggregations such as seasonal, monthly, weekly and as an example both “Daily” weather forecast fields (e.g., days 1-16) or “Hourly” weather forecasts (e.g., model hours 1-384). See FIG. 5. [0132] a. ROLLING [0133] i. KORD MR0! (Model Run Now) is the current model (whatever wx model is running now, or the last model to run (if in between model runs). This instruction extracts the latest weather model which is coming in now or the last model to have run. [0134] 1. EX: KORD MR0!-GFS [0135] ii. KORD MR0!_12z-GFS is requesting the latest 12z model run. [0136] 1. To access any previous model run of the 12z see below. If the 12z was omitted the symbol would return the last run of the GFS, whichever model run was the most current. [0137] iii. MR1! (1 through 99) is that many “model runs ago” for whatever comes after it. Will go back x #of runs for the suffix. So MR1!_12x-GFS will be yesterday's 12z GFS, MR1! GFS will be 1 model run ago. [0138] 1. KORD MR1!-GFS (for-1 model run back, from which ever model run of the GFS is current) [0139] 2. KORD MR1!_12z-GFS (for the previous days run of the specific 12z GFS) [0140] b. Specific [0141] i. Daily [0142] 1. KORD MR200301-GFS (YY,MM,DD): returns all the GFS weather forecast model runs which were produced on Mar. 1, 2020 (aka Model Run Date) This function provides extraction of a historical weather forecast from specific “model run dates” and for specific weather models and model runs. [0143] ii. Model Run Specific [0144] 1. KORD MR200301_12z-GFS (YY,MM,DD): returns only the 1 model run (i.e. 12z) of the GFS which was produced on Mar. 1, 2020 and is the 12z. This function provides extraction of a historical weather forecast from specific “model run dates” and for specific weather models and runs. [0145] 3) Progression (aka Forecast Dated Day & Forecast Dated Hour): Provides an analysis of how the forecast models changed over time for a specific Forecast Valid ‘Day’, ‘Date’ or ‘time of day’ (aka hour) on a specific Date also known as the “Target Date. Can be displayed in “Exchange time” or “interval start”). Note: when charting these symbols it may be useful to ‘chart’ using “exchange time” (on the x axis) to view this progression data in the correct order (i.e. as it was released by model run), considering the Forecast Valid ‘day/date’ is locked into the symbol (i.e. the Forecast Valid data is normally plotted on the x axis using local day/date. See FIGS. 6A-6C. [0146] a. Specific: [0147] i. Daily Forecast: KORD FDD200301_12z-GFS (YY,MM,DD): This symbol provides a history of all the 12z GFS weather forecasts, which pertain to that specific target ‘Date’ which is noted in the symbol; e.g., Mar. 1, 2020. Note: If the “12z” was omitted in this symbol, the alternate symbol design would return “all model runs” of the GFS, which provided a (daily) forecast for that specific date. [0148] ii. Hourly Forecast: KORD FDH20030113_12z-GFS (YY,MM,DD,HH): Same as above, but provides a specific hour (aka time of day) on a specific date. This symbol provides a history of all the 12z GFS Forecasts and may return the weather forecasts which pertain to that specific ‘Date’, e.g., Mar. 1, 2020 at 1 pm “local time” for that specific location. In an example, this specific “time of day” is noted by hour “13”, which is appended to the date. The “13” represents the “local time” of day, for that specific location. The last 2 digits range from 01 (1 am) to 24 (midnight). Note: If the “12z” was omitted in this symbol, the new symbol would return “all model runs” of the GFS, which provided an “Hourly” forecast for that time of day, for that specific date. [0149] b. Rolling [0150] i. Daily Forecast: KORD FDD2!-12z-GFS: This symbol acts exactly in the same manner as above, except that the specific “Date” may be unknown (just the Forecast Valid “Day” may be known). For example if a user did not know tomorrow's “Date” this rolling symbology structure may be used to lock into the “Date” which is now “Day2”. [0151] 1. In summary: This symbol provides a history of all the 12z GFS weather forecasts, which pertain to that specific target “Date” which is now noted in the symbol; as “Day2”. Tomorrow that same “Day2” element in the weather symbol, will refer to a different specific “Forecast valid Date”. [0152] 2. Note: If the “12z” was

omitted in this symbol, the new symbol would return “all 4 model runs” of the GFS, which provided a daily forecast for that specific ‘Date’, which is currently Day 2. [0153] 3. Note because “Day 1” equals “Today”. That is different than the manner in which we normally count. For example, FDD7! is 6 days from today. [0154] ii. Hourly Forecast: KORD FDD2_H13!-12z-GFS: Provides a history of all the 12z Forecasts and may return the weather forecasts which pertain to that specific “Forecast Day” at that specific “Time of Day” on tomorrow Day 2, being “Local Hour 13” which is 1 pm. Note: These times are expressed as the local hour of the specific location's day/date. [0155] 1. If the “12z” was omitted in this symbol, the new symbol would return “all 4 model runs” of the GFS, which provided a Hourly forecast for that specific date & time which is currently defined as Model Hour 120. [0156] 2. Note the H is two (2) digits ranging from 1-24, with 1=1 am and 24=midnight. [0157] iii. Model Hour: KORD FDH012z-GFS: Returns the history of 12z GFS weather forecasts for only model hour 0. This is independent of what every the local day/time is related to the forecast Valid period (acting very similar to an accuracy workflow like the KORD FCH13!_12z-GFS example (below). [0158] 1. If the “12z” was omitted in this symbol, the new symbol would return “all 4 model runs” of the GFS, which provided a Hourly forecast for Model Hour 0. [0159] 2. Note that His two (2) digits ranging from 0-384. [0160] 4) Accuracy (aka Forecast Continuation Day & Forecast Continuation Hour): This symbology structure may line up the weather forecast data in a manner which allows a user to easily match the extracted weather forecasts with weather observations. These observations can then be subtracted from the extracted forecasts to derive forecast accuracy statistics such as Bias, root mean square error (RMSE), Error, standard deviation (std) of error, etc. In summary, this Accuracy based symbology may return all historical weather forecast information in a manner which lines up all previous weather forecasts for a specific (rolling) days in the future (like all the ‘Day 1’ forecasts would be lined up (then all day 2, 3, 4, etc.) or specific (rolling) “model hours” in the future (like model hour 3, 6, 9, 12, etc. . . .) See FIGS. 7A and 7B. [0161] a. Rolling [0162] i. Daily Forecast: KORD FCD1!_12z-GFS: Provides the “Day 1” forecast from all 12z GFS weather forecasts stored in a database. [0163] 1. Note: If the “12z” was omitted in this symbol, the new symbol would return “all 4 model runs” of the GFS. [0164] ii. Hourly Forecast: KORD FCH13!_12z-GFS: Provides the “model hour 13” forecast from all 12z GFS weather forecasts stored in a database. Note: These are forecast “model hours” and range from 0-384 as opposed to the manner in which “Forecast Valid” date & time of day is used (as in FDH above) which starts at ‘1’ indicating 1 am local time, for a specific location. [0165] 1. Note: If the “12z” was omitted in this symbol, the new symbol would return “all 4 model runs” of the GFS. [0166] 5) METAR SYMBOLOGY: METAR is one particular source of global weather observations. METAR reports weather observations from key airports (globally) in a relatively timely manner, so there is typically little (if any) cleaning to this weather observation data stream. [0167] a. General Structure <Location><Suffix> [0168] i. KORD-METR [0169] 6) MISCELLANEOUS (MISC.): [0170] a. AutoListing: save the user time from typing in many different symbols. [0171] i. MR's [0172] 1. *<Location> MR! <Model>-returns 4 current model runs [0173] 2. Example: *KORD MR!-GFS [0174] 3. *<Location> MR!<Model><N-N>-returns the last N Model Runs [0175] 4. Example: *KORD MR!_12z-GFS 0-2 [0176] ii. FDD's (Progression) [0177] 1. *<Location> FDD!<Model>-returns FDD 1-16 [0178] 2. Example: *KORD FDD!_12z-GFS [0179] 3. *<Location> FDD!<Model><N-N>-returns the last N FDD [0180] 4. Example: *KORD FDD!_12z-GFS 1-5 [0181] 5. *<Location> FDH!<Model>-returns all hourly symbols [0182] 6. Example: *KORD FDH!_12z [0183] 7. *<Location> FDH!<Model><N-N>-returns N hourly symbols [0184] 8. Example: *KORD FDH!_12z 1-12 [0185] iii. FC's (Accuracy) [0186] 1. *<Location> FCD!<Model>-returns all continuous daily history symbols [0187] 2. Example: *KORD FCD!_12z [0188] 3. *<Location> FCD!<Model><N-N>-returns N continuous daily history [0189] 4. Example: *KORD FCD!_12z 1-12 [0190] b. Forecast Valid Days: 1-16 (or beyond, depending upon the duration of the various wx forecast models): Day 1=today (there is no Day 0). In some examples, “Day 1=today”, so “Forecast Valid Day 4” is “3

days” from now (For example: (1) today/Monday, (2) tomorrow/Tuesday, (3) next day/Wednesday, (4) target day/Thursday). [0191] c. Model Runs (00z-23z). Most Weather Forecast Models are 00z, 06z, 12z, 18z) [0192] d. Forecast Valid Hours in a Specific ‘Date’ or ‘Day’ (e.g., in local time) [0193] i. Range from 1-24 (e.g., 13=1:00 pm local time) [0194] e. Forecast ‘Model Hours’, in some examples, may be from the start of the weather model run and extend out into the future. In some examples, weather forecast models are 384 or 240 hours into the future. [0195] i. Range from 0-384 (for example). [0196] f. Ensemble Variables [0197] i. Location_ {underscore} Ensemble variable {space} Function_ {underscore} Model Run {Dash}-Wx Model [0198] 1. EX: KORD_MEAN MR0!_12z-GFS [0199] ii. In some examples, a default variable for Ensemble models is the “MEAN” or average of all perturbations [0200] iii. Other selectable Ensemble variables may include: [0201] 1. MAX (highest ensemble member) [0202] 2. MIN (lowest ensemble member) [0203] 3. MEAN (Average of all ensemble members or also can be used as 50% of all members) [0204] 4. +1/-1 STD (1 standard deviation is derived which can be added or subtract from the MEAN) [0205] 5. PER10 (10.sup.th percentile of all members) [0206] 6. PER25 (25.sup.th percentile of all members) [0207] 7. PER75 (75.sup.th percentile of all members) [0208] 8. PER90 (90.sup.th percentile of all members) [0209] iv. GFS ENS: Fast vs Full— [0210] 1. Fast may be derived using, for example, 50% of the total Ensemble members. For example, the first 10, of the 20 ensemble members. These preliminary values have an extraordinarily high correlation to capture the directional change (and magnitude) of the Full forecast. The benefit to providing this directional perspective is receiving this information approximately 30 seconds before all ensemble members can be modeled and transmitted to other users. This concept works for streaming Ensemble models (e.g., GFS ENS), when ALL Ensemble members are computed. [0211] v. Interpolated Values: [0212] 1. In some examples systems of the present disclosure may receive weather forecasts (which may arrive “As-Is”), and interpolate the forecast hours which were not forecasted by the weather model. In some examples, systems of the present disclosure may extract the “Trailing Minimum” and “Trailing Maximum” temperatures from the weather forecast grib files, and then using a diurnal pattern based upon data (or a blend of data), may place these trailing Minimum & Maximum values between the reported forecasts which reside on the top of each hour. [0213] vi. % Complete: [0214] 1. This is a unique element of the present disclosure which allows users an advantage, viewing official daily weather forecast information a few minutes prior to the rest of the marketplace. This process may provide “preliminary forecasted values” for daily Max, Min, Avg temperature, or daily GWDD, Heating Degree Day (HDD), Cooling Degree Day (CDD) values, when the weather forecast model is partially (e.g., 50%) completed for any forecast Day and provide real-time updates as that forecast Day runs through 100% complete. The preliminary Daily values of the present disclosure may provide users insight, a few minutes in advance of other services, who only publish a forecast Day at 100%.

[0215] FIGS. 8A and 8B provide for specific operational examples of how the various weather symbology structures operate using sample weather forecasts. For example, FIG. 8A illustrates example interactive GUI 800 of a spreadsheet application, illustrating examples symbology construction, according to an aspect of the present disclosure. FIG. 8B illustrates example interactive GUI 810 illustrating examples of various weather symbology structures for a spreadsheet application.

[0216] Table 3 provides brief descriptions of example weather terminology used throughout the present disclosure.

TABLE-US-00003 TABLE 3 Example Weather Symbology Terminology

Term	Description
Receipt Time	(e.g., The time reported is the time at which the file arrived at the server(s). exchange time)
Location	Locations may be global airport codes (ICAO) and/or custom regions designed to represent specific geographic regions related to a specific commodity (such as a gas weighted Energy Information Administration (EIA) region-based commodity or one or more Independent System Operator (ISO) regions to mimic an entity that controls and monitors the operation of an

electrical power system for a given geographic region). Example Weather GFS- GFS (¼ degree spatial resolution. Time steps: mhr 0-120 @ 1 hr; Models/Model mhr 121-384 @ 3 hr - run 4x per day). Names GEFS - GFS Ensemble (¼ degree spatial resolution. Time steps: mhr 0- 120 @ 1 hr - mhr 121-384 @ 3 hr. run 4x per day. Fast = 17 members/ Full = 35 members. ECM- ECMWF's high resolution model (¼ degree spatial resolution for surface variables and 1 degree resolution for upper air variables. Time steps: mhr 0--90@ 1 hr_ mhr 93-144 @3 hr _ mhr 150-240 @6 hr _ 0z & 12z ONLY. 06 & 18z mhr 0-90 @ 1 hr. ECE - ECMWF Ensemble model (ENS) (½ degree spatial resolution for surface variables and 1 degree resolution for upper air variables. Time steps: mhr 0--90@ 1 hr _ mhr 93-144 @3 hr _ mhr 150-240 @6 hr _ 0z & 12z ONLY. 06 & 18z mhr 0-90 @ 1 hr_ mhr 93-144 @3 hr. Full = 52 members. ECEX- ECMWF Ensemble Extension (ENS) (½ degree spatial resolution for surface variables and 1 degree resolution for upper air variables. Time steps: mhr 0--384 @24 hr - run 2x per week). GEFSX - GFS Ensemble EXTENSION (¼ degree spatial resolution. Time steps: mhr 0-120 @ 1 hr - mhr 121-384 @3 hr. run 1x per day. Fast = 11 members/Full = 22 members. CFSX - CFS Ensemble EXTENSION (¼ degree spatial resolution. Time steps: mhr 0-5,400 (9 mo) @6 hr -. run 1x per day. Full = 5 members. SEAS - ECMWF's Seasonal Model (¼ degree spatial resolution. Time steps: mhr 0-5,040 (7 mo) @6 hr -. run 5th of every month. Full = 5 members. Note that the spatial resolution is subject to change based upon the sources. Model Run Currently 0z, 6z, 12z, 18z. This may change depending upon modification by weather agencies to modify the frequency of model runs. Model Hour (aka Currently integers between 0-384. 0 hour is the initial hour from which Forecast Hour) the weather forecast is initialized and the model physics then prognosticates forward into the future. Time Steps/ Is the difference between Model Hours described above. For example if Reporting interval the forecasted model hours were 0, 3, 6, 9, 12, etc., the model would possess '3 hour' time steps. Forecast Valid A particular time for which a weather forecast will pertain to a calendar Date 'Date'. Dates are based upon midnight-to-midnight local time of the forecasted location. "FV Dates" are mostly used in "Specific Dates" for the weather symbology structure. Forecast Valid A particular time for which a weather forecast will pertain to a calendar Day 'Day' in the future, which is based upon midnight-to-midnight local time of the forecasted location. Mostly used in "Rolling Dates" for the weather symbology structure. For example; Day 2 represents a forecast for tomorrow, regardless of which specific calendar date it represents. Forecast Valid A particular weather forecast for a particular time of Day or Date, which Hour (Time of is based upon midnight-to-midnight local time of the forecasted location. Day) This Forecast Valid Hour can be applied in either "specific" or rolling" weather symbology structure. For example; HH01 is 1:00 am in the morning, local time. How Model Hour Each Model Hour provides a global forecast which may correspond to a relates to Forecast specific Forecast Valid Day, Date and Time, given the global time zone Valid Day/FV of the specific forecast location. For example, Model Hour 1 may be Date/FV Time of equal to 1am London time, but may directly correspond to 7pm Chicago Day time (which is 6 hrs behind London). (these are FV Day = Day 1, 2, 3, etc . . . expressed in local FV Date = a specific calendar date time for the FV Time of Day = a specific time on a specific Date forecast location) Percent Complete AKA "Preliminary Daily Values" provides users an early view of what the forecasted daily value will (likely) be a few minutes in advance. Percent Complete displays the percentage of potential hourly data used to derive a daily value for a specific forecast date/day. For example, the second forecast day (for some location) consists of 24 forecast hours. When 12 of those 24 files are received, a preliminary daily value may be generated for that specific day, As each subsequent forecast hour updates from the forecast model, so will the % complete, unit 100% is achieved for that specific forecast day. "As-Is" model This refers to the specific model hours in which a weather forecast model data (aka "Raw" provides a forecast. model data) Interpolated This refers to the forecast hours in between the "As-Is model data" from which the weather forecast model did not provide weather forecast data. The present disclosure implements a sophisticated method to interpolate those in between hours. Blended Indicates whether a weather forecast data is combined

with observation data. Ensemble Models Models runs may produce eight (8) distinct outputs regarding ensemble statistics for a given variable. Max - The highest ensemble member Min - The lowest ensemble member Mean - The average of all ensemble members. Std - 1 standard deviation of all the ensemble members. 10% - - 10.sup.th percentile of the ensemble values. 25% - - 25.sup.th percentile of the ensemble values. 50% - - 50.sup.th percentile of the ensemble values can be displayed by using the “Mean” (listed above). 75% - - 75.sup.th percentile of the ensemble values. 90% - - 90.sup.th percentile of the ensemble values. Ensemble Member This is how many ensemble members are included in the analysis. For Count GEFS ‘Fast’, an early look at the ensemble data may be provided using a portion (e.g., 22) of the total (e.g., 45) GEFS ensemble members. Guidance Charts Plot weather forecasts going out into the future (standard weather charts) Progression Charts May “lock” a particular “Forecast Valid Date or Time of Day” (also known as the “Target Date”) and show how the previous weather forecasts have changed over time for that “Forecast Valid” Day, Date or Time of Day. Target Date Used in a progression chart it is the specific Date and/or Time of Day which is held constant from which an examination of how all the previous weather forecast Models/Model Runs have changed (overtime) for that particular Day/Date/Time. Alpha Charts This is the unique display in the marketplace and exclusive to aspects of the present disclosure. Streaming weather forecasts as they are made available into the public domain in conjunction with real-time market prices, can more easily expose the relationship between changing weather perspectives and changes in traded instruments. NOAA National Oceanic and Atmospheric Administration www.noaa.gov NCEP National Center for Environmental prediction www.ncep.noaa.gov ECMWF European Center of Medium Range Weather Forecasts www.ecmwf.int CFSR A source of global weather observations. The Re-Analysis data from the CFS weather forecast model may be used to compile, in some examples, greater than 41 analog years and at least four climatological datasets. METAR A source of streaming global weather observations from instrumentation located at airports. Exchange Time Is the time when the data is received. (aka receipt time) Interval Start Time A setting used in charts to allow for Future dates, so that weather forecast data may be plotted forward in time.

Example: Weather Data with Respect to Exchange Time and Price Data

[0217] The following is an example of charting weather data with respect to exchange time and price data. FIG. 4A is screenshot including a chart of example weather forecast data and price data as a function of exchange time. The example shown in FIG. 4A illustrates a scenario where each weather Forecast Valid Day is being produced by the weather forecast model (source). The x axis describes the “exchange time” which each dot (elements **402**, **404**) is received. Each dot **402** is a ‘preliminary view’ of that day's forecast element (such as daily average temperature, GWDD, HDD, CDD, etc.), in which a condition code (“% complete”) may begin providing ‘preliminary views’ when a forecast day is 50% (for example) and may stop when that forecast day is completed at 100%. Dot **404** at the end of a forecast valid day is the 100% conclusion.

[0218] Not shown here, is the price of any traded commodity using a minute Open/High/Low/Close (OHLC) bar. In general, OHLC represents an example manner in which traded prices may be displayed, over some time interval (e.g., one minute, five minutes, fifteen minutes, one day, etc.). For example, trading data may be organized into one minute intervals and a single bar may depict the OHLC over each of the one minute time spans (or any suitable time period). In some examples, one minute OHLC bars may be displayed together with weather “tick” data to aid in identifying trading opportunities. The presentation of the combination of the two data sets on interactive GUI **120** provides advantages, including in makes it easy to visualize and track (including in real-time) when weather moves in a significant manner to move the price of a traded financial instrument.

[0219] FIGS. 4B-4D are screenshots of an example 16 day weather forecast for a particular location as a function of exchange time with respect to natural gas price data. In particular, FIGS. 4B and 4C show an example 16 day weather forecast from the 12z GFS weather forecast model in

‘exchange time’ for the GWDD variable as it relates to the price of Natural gas at Henry Hub. FIG. 4D is a screenshot of an example weather alpha workflow chart as a function of exchange time. As shown in FIG. 4D, when GWDD goes down, price should go down and vice-a-versa.

Example of a Guidance Chart

[0220] In some examples, the weather symbology of the present disclosure may be used for guidance weather perspective workflows, and may be configured to generate guidance charts of weather forecast datasets. FIG. 5 is a screenshot including an example guidance chart associated with a guidance weather workflow, the guidance chart illustrating weather forecast data as a function of a forward time period. The example chart starts ‘now’ at the current hour and extends forward in time 16 days (hour-by hour or 0-384 hours into the future) as seen on the x axis and Temperature displayed (in F) on the y axis for Chicago O’Hare airport. Set of curves 502 shows the 0z GEFS Ensemble mean (curve 502-1) as well as the +1 STD (curve 502-2) and -1 STD (curve 502-2) around the mean (curve 502-1). Set of curves 504 shows the 0z ECE Ensemble mean (curve 504-1) as well as the +1 STD (curve 504-2) and -1 STD (curve 504-2) around the mean (curve 504-1).

[0221] Table 4 is an example output of the data behind the same guidance chart with the weather symbology. In some examples, system 102 may provide users with the ability to plot this data in various presentation formats (e.g., a data grid, a spreadsheet, a watchlist) and may update the resulting data in real-time as the underlying data input(s) change.

TABLE-US-00004 TABLE 4 Example Weather Chart Output Data 0Z +1STD -1STD -1STD +1STD

Date	Time	GEFS 0z	GEFS 0z	ECE 0z	GEFS 0z	ECE 0z	ECE 0z
Feb. 21, 2020	22:00	24.4	37.3	11.9	28.1	48.5	Feb. 21, 2020 21:00 24.3 36.8 38.9 11.9 28.7 49
Feb. 21, 2020	20:00	24.2	36.6	11.8	29.2	49.6	Feb. 21, 2020 19:00 24.1 36.5 40 11.8 29.7 50.2
Feb. 21, 2020	18:00	24	36.3	11.8	30.1	50.9	Feb. 21, 2020 17:00 23 35.4 38.1 10.6 28.1 48.2
Feb. 21, 2020	16:00	22	34.4	9.5	25.8	45.6	Feb. 21, 2020 15:00 20.9 33.5 33.4 8.3 23.5 43.2
Feb. 21, 2020	14:00	19.9	32.6	7.1	21.1	40.8	Feb. 21, 2020 13:00 18.8 31.8 28.6 5.9 18.6 38.6
Feb. 21, 2020	12:00	17.8	31	4.7	15.9	36.5	Feb. 21, 2020 11:00 17.9 30.8 26.5 5 16.4 36.5
Feb. 21, 2020	10:00	18	30.7	16.9	36.5	Feb. 21, 2020 9:00 18.2 30.6 26.9 5.7 17.3 36.6	
Feb. 21, 2020	8:00	18.3	30.6	17.7	36.7	Feb. 21, 2020 7:00 18.4 30.6 27.4 6.2 17.9 36.9	
Feb. 21, 2020	6:00	18.5	30.6	18.2	37.1	Feb. 21, 2020 5:00 18.7 30.5 28.5 6.9 19.1 37.8	
Feb. 21, 2020	4:00	18.9	30.5	20	38.6	Feb. 21, 2020 3:00 19.1 30.6 30.1 7.7 20.8 39.4	
Feb. 21, 2020	2:00	19.3	30.6	21.5	40.3	Feb. 21, 2020 1:00 19.6 30.7 31.7 8.4 22.1 41.3	
Feb. 21, 2020	0:00	19.8	30.9	22.7	42.4	Feb. 20, 2020 23:00 20.1 31 33.1 9.3 23.3 42.9	
Feb. 20, 2020	22:00	20.5	31.1	23.8	43.6	Feb. 20, 2020 21:00 20.8 31.3 34.3 10.4 24.3 44.2	
Feb. 20, 2020	20:00	21.2	31.5	10.8	24.8	44.9	Feb. 20, 2020 19:00 21.5 31.8 35.4 11.2 25.2 45.7

Example: Progression Charts

[0222] In some examples, the weather symbology of the present disclosure may be used for progression weather perspective workflows, and may be configured to generate progression charts of weather forecast datasets. FIGS. 6A-6C show the convergence of weather forecast models, as the forecast changes over time, from day-to-day for a specific Forecast Valid Day, using Progression charts of the present disclosure. In particular, FIG. 6A is a screenshot including an example progression chart associated with a progression weather workflow; FIG. 6B is a screenshot of an example progression chart illustrating natural gas storage estimates and weather model progression as a function of forecast date; and FIG. 6C is an example progression chart for several weather models as a function of forecast date.

[0223] In FIG. 6A, as shown in the far left (16 days ago), there was more risk for today's forecast (Friday February 7.sup.th), which appeared to flatten out approximately 5 days prior to today February 7.sup.th (yet the models still show a few degrees difference). FIG. 6B, illustrates example natural gas storage estimates (reads in the opposite direction of FIG. 6A, from right to left).

Example: Accuracy Charts

[0224] In some examples, the weather symbology of the present disclosure may be used for accuracy weather perspective workflows, and may be configured to generate accuracy charts of weather forecast datasets. FIG. 7A is an example initiation screen for generating an accuracy chart. In some examples, the weather symbology may easily organize weather forecast model data using the FCD and FCH commands, allowing weather forecast accuracy to be quickly understood via presentation on interactive GUI 120. It may be appreciated that any knowledge of which weather forecast is most likely to be correct may be an informational advantage, particularly for interacting with market data.

[0225] FIG. 7B is an example accuracy chart associated with an accuracy weather workflow. This data display shown in FIG. 7B is an example using the weather symbol structure of the present disclosure using the Accuracy style codes which begin with FCD.

Example: System Operation

[0226] An example of system operation of weather integration server 116 is provided below. In this example, MR refers to a Model Run, MRD refers to a Model Run Date, MRH refers to a Model Run Hour, MRO refers to a Model Run Offset, FDD refers to a Forecast Dated Date, DD refers to a Dated Date, DDO refers to a dated date offset, FCD refers to a Forecast Continuation Day, CDO refers to a Continuation Day Offset, FDH refers to a Forecast Dated Hour, DH refers to a Dated Hour, DHO refers to a Dated Hour Offset, FCH refers to a Forecast Continuation Hour, CHO refers to a Continuation Hour Offset., "LOC" refers to location and "REQ" refers to request. In Table, 5, DM represents a data manager (such as for data cache(s) 124).

[0227] Table 5 below illustrates non-limiting examples of system operation according to aspects of the present disclosure.

TABLE-US-00005 TABLE 5 Examples of System Operation Request Request Type
Rolling/Specific DM Underlying Symbol KBOS MR0!-GFS MR Rolling >Return KBOS Latest Model Run >MRD is REF_DATE: current MR190316_6z-GFS for Boston from calculated 2019 Mar. 16 day's latest Key gets updated GFS from (Today) Forecast every model run. MODEL: GFS Reference data. User may be updated LOC: KBOS date (Today), >Keeps user every model run. REQ: MR Number of updated of Chart user may clear MRO: 0 Runs in the new model down and re-request day and MRO. runs. historical data every >NO MRH - >Confirmed new model run. use latest clear down Users may receive model run for all model data updates (hourly/ hour. run changes. daily) as new data comes in from data feed. Underlying symbols are updated on model run changes. Ex for GFS: 1. KBOS MRYMMDD_0z-GF 2. KBOS MRYMMDD_6z-GF 3. KBOS MRYMMDD_12z-GF 4. KBOS MRYMMDD_18z-GF KBOS MR1!-GFS MR Rolling >DM may >Expired Symbol. Latest Model Run >MRD is REF_DATE: return >No Underlying symbol. (01) for Boston calculated 2019 Mar. 16 expired >User may be updated from GFS from (Today) symbol. every model run. MODEL: GFS Reference MRO: -1 >User may >Chart user may clear LOC: KBOS date (Today), 1 run back get updates down and re request REQ: MR Number of on current historical data every new MRO: -1 Runs in the model runs model run. day and MRO. for symbol >NO MRH - and source. use latest >Live Chart model run request may hour. generate new historical request for each model run update. KBOS MR0!_00z-GFS MR Rolling >If current KBOS MR 190316_0z-GFS Latest 00z for MRD is REF_DATE: day has >User may be updated Boston from GFS calculated 2019 Mar. 16 MRH: 0Z every model run and MODEL: GFS from (Today) DM may requested MRH. LOC: KBOS Reference return latest >Chart user may clear REQ: MR day(Today), forecast data down and re-request MRO: 0 Number of for today historical data every new MRH: 00z Runs in the with MRH model run. day and MRO. (00z). >Else may return expired symbol. KBOS MR1!_0Z-GFS MR Rolling >DM may >Expired Symbol, Last Model Hour MRD can be REF_DATE: return >No Underlying symbol 00z calculated 2019 Mar. 16 expired >User may be updated for Boston from from reference (Today) symbol. every model run and GFS day(Today), >User may requested MRH. MODEL: GFS Number of receive >Chart user may clear LOC: KBOS Runs in the updates on down and re request REQ: MR day, MRO and

current historical data every new MRO: -1 MRH. model runs model run. MRH: 00z for symbol and source. >Live Chart request's may generate new historical request for each model run update and MRH. KBOS MR Specific >DM may KBOS MR190316_0z-GFS MR190316_0z-GFS MRD is from MRD: return data if Same as client symbol Return Model run: request Mar. 16, 2019 current day 2019 Mar. 16 at (Use format to MRH = 00z is specific Model run: 00z determine day and has data for Boston specific) date for from MRH (00z) MODEL: GFS And no LOC: KBOS further REQ: MR updates. MRD: Mar. 16, 2019 >if MRD is MRH: 00z current day but MRH is in a future DM may wait until that MRH is received in data feed. KBOS MR MR Specific >DM may KBOS MR 190316_6z-GFS 190316_6z-GFS MRD is from MRD: return data if Client symbol Return Model run: request Mar. 16, 2019 current day 2019 Mar. 16 at (Use format to MRH = 06z is specific Model run: 00z determine day and has data for Boston specific) date for from MRH (00z) MODEL: GFS and no LOC: KBOS further REQ: MR updates. MRD: Mar. 16, 2019 >if MRD is MRH: 06z current day but MRH is in future DM may wait until that MRH received in data feed. KBOS FDD8!-GFS FDD Rolling >If current KBOS FDD190323-GFS 7 h Forecast Dated Forecast REF_DATE: day has the Key gets updated Day from Dated Date is 2019 Mar. 16 interested every day. Today(Current calculated (Today) forecast User may be updated day: 1) from date, DM every model run. MODEL: GFS Reference may return Chart user may clear LOC: KBOS Date and the latest down and re request REQ: FDD DDO forecast data historical data if DDO: 8 Ex: for that date. needed. Today: >Keeps user 2019 Mar. 16 updated for DDO: 8 days new runs (note current which have day: 1) the FDD. Interested Date: 2019 Mar. 23 Database Options: Find all Model Run Dates (or time indexed field) which may have particular FDD. Or use an index for Forecast Date to find Data. KBOS FDD Specific >May return KBOS FDD190321-GFS FDD190321-GFS FDD is from FDD: expired Client symbol Data from all request. 2019 Mar. 16 symbol if Model runs with Database FDD is not forecast for Options: Find current day. 2019 Mar. 21 all Model Run >Subscribes MODEL: GFS Dates (or time and Returns LOC: KBOS indexed field) data if FDD REQ: FDD which may is Current FDD: 2019 Mar. 16 have Date and in particular future. FDD. >No further Or use an updates after index for FDD goes Forecast Date out of scope. to find Data. KBOS FDD Rolling >If current KBOS FDD12!_00z-GFS Forecast REF_DATE: day has the FDD190327_00z-GFS 12.sup.th Forecast Dated Dated Date is 2019 Mar. 16 interested Key gets updated Day from Today calculated (Today) FDD and every day. with Model Run from MRH, DM User may be updated Hour of 00z. Reference may return every model run and i.e., 00z runs with Date and the latest requested MRH. forecast date of DDO. forecast data Chart user may clear 2019 Mar. 27 Ex: for that date. down and re request MODEL: GFS Today: >Keeps user historical data if LOC: KBOS 2019 Mar. 16 updated for needed. REQ: FDD DDO: 12 days new runs DDO: 12 Interested which have MRH: 00z Date: the FDD and 2019 Mar. 27 MRH. Database Options: Find all Model Run Dates (or time indexed field) which may have particular FDD. Or use an index for Forecast Date to find Data. KBOS FCD1!-GFS FCD Rolling >Subscribes >No Underlying symbol. Forecast Day 1 >Use the REF_DATE: and Returns >User may be updated (from possible total Model Run 2019 Mar. 16 data for all every model run. 16 (1-17)) Model Limit and (Today) model runs Runs filter with for that MODEL: GFS FCD. Forecast LOC: KBOS >Confirm max Day (1-17). REQ: FCD 'n' model runs >First CDO: 1 limit with Watchlist Model Run Limit: DM. record Max Model Runs, returned going back to n may be model runs latest update for current day. KBOS FCD Rolling >Subscribes >No Underlying symbol FCD1!_06z-GFS >Use the REF_DATE: and Returns >User may be updated Forecast Day 1 Model Run 2019 Mar. 16 data for all every model run and (from possible total Limit and (Today) model runs requested MRH. 16) Model Runs filter with for that MODEL: GFS FCD and Forecast LOC: KBOS MRH. Day (1-17) REQ: FCD >Confirm max with MRH CDO: 1 'n' model runs of 06Z MRH: 06z limit with >Watchlist Model Run Limit: DM. may be Max Model Runs, empty if going back to n there is no model runs record for FCD and MRH. >But DM may update after referenced Forecast Day comes in a Model Run. Example: Model Run Updates to Users

[0228] In some examples, weather integration server **116** may provide one or more model run updates in response to latest model run requests and/or rolling back requests from users (e.g., via user device(s) **108**).

[0229] For a Latest Model Run (MR0) Request, a model run update may be sent for every new model run for the related source and location. In some examples, the model run update may be used by the chart user to clear a history, such that no new history request may be needed.

[0230] For a Rolling Back (MRNN<1-99>) Request, in some examples, when a user subscribes for a model run going backwards, an expired symbol response may be provided (or a different weather response indicating the information is historical). From there on forward, the user may be provided with updates for every new model run for that location and source.

[0231] In some examples, an expired symbol response and model run update may be used by all users (e.g., data may get updates for individual fields) to clear state and chart users re-request historical data from one or more databases.

[0232] In some examples, when a new model run starts, all chart users may need the chart to be re-drawn (which may cause weather integration server **116** to receive a request for historical data (if not cached) for a previous model run). To avoid all chart users requesting historical data at a same time during new model runs, weather integration server **116** may include logic to pre-request data at a pre-defined user interval before a new run starts.

[0233] In some examples, for any rolling request if the user request includes a model run hour, the user may be notified when that specific model run hour is received from a weather data source(s) data feed.

Example: Forecast Dated Date Requests (FDD) and Forecast Continuation Day (FCD)

[0234] In some examples FDD requests, FDH requests and/or FCD requests may be received that may be subscribed to via a previous model run, a current model run and/or for a future model run. FDD and FDH requests relate to the progression weather perspective, and FCH requests relate to the accuracy weather perspective (described above).

[0235] For example, when a user requests data for FDD and FCD and are subscribed to current model run, the resulting chart(s) may be updated for every new model run for the related source and location. In some examples, for every new day runs (e.g., 00z), all users with charts open may have charts re-drawn for FDD/FDH and may request historical data for that FDD/FDH and FCD/FCH.

[0236] For future model runs, a user may subscribe for a future symbol. In this example, the user may receive no forecast data response and may receive an update on that symbol when the future event occurs.

[0237] In some examples, users may not need to re-request historical data for new model run updates.

Example: Weather Field Filtering

[0238] In some examples, a filter option may be provided in weather impact dashboard **122**. The filter option may provide users (e.g., via user device(s) **108**) the ability to include a filter option to a weather impact request. In one example, the format for a filter option may be:

filters="field_1=<value>&field_2=<value>"

For example, for filtering by daily symbol (interval_type=D) and with 100% complete, the following filters option may be used: filters="interval_type=D&PercentComplete=100", where the term "interval_type" may include Hourly (H, default) and Daily (D). In some examples, if the interval_type is not specified, system **102** may lookup interval_type after inspecting the fields. In some examples, an additional optional filter DEBUG=T&columns=* may be used to request all fields.

Example: Model Run Information

[0239] In one non-limiting example, weather integration server **116** may provide model run

information for GFS and GEFS weather data models. The model run information may include model hours (e.g., 0z, 6z, 12z, 18z, based on GMT), may provide about 385 hourly records and about 209 daily records.

Example: Symbology Specification

[0240] Next, an example symbology specification is described, according to an aspect of the present disclosure.

[0241] In this non-limiting example, the symbology for forecasts may include at least five different methods to retrieve a symbol and/or data. Example retrieval methods may include: (i) by symbol type and field, (ii) by symbol type and aggregation, (iii) by model run and model run date, (iv) by valid day/date-time and (v) by valid day or date. FIG. 8A illustrates an example user interface illustrating examples of symbology construction.

[0242] Table 6 illustrates an example symbol list and available fields that may be used for retrieval by symbol type and field. Table 7 illustrates an example symbol list and available aggregations that may be used for retrieval by symbol type and aggregation.

TABLE-US-00006	TABLE 6	Example Symbol List and Available Fields	Symbol Type	Example
Intraday	Daily	Model Run - Rolling KORD MR0-GFS	Yes	Yes
Model Run - Specific	KORD MR190228_12z-	Yes	Yes	GFS
Model Run - Rolling	run KORD MR0_0z-GFS	Yes	Yes	Horizon
Day - Rolling	KORD FH13-GFS	No	Yes	Horizon
Hour - Rolling	KORD FHH1-GFS	Yes	No	Horizon
Day - Specific	KORD FH190228-GFS	No	Yes	Horizon
Hour - Specific	KORD FH19022812-GFS	Yes	No	Horizon
Day - Rolling	run KORD FH12_12z-GFS	No	Yes	Horizon
Hour - Rolling	run KORD FHH1_18z-GFS	Yes	No	Valid Day
Day - Rolling	KORD FD1-GFS	No	Yes	Valid Day
Day - Rolling	run KORD FD1_12z-GFS	No	Yes	

TABLE-US-00007	TABLE 7	Example Symbol List and Available Aggregations	Symbol Type	Example
Intraday?	Daily?	Model Run - Rolling KORD MR0-GFS	Yes (hourly)	Yes
Model Run - Specific	KORD MR190228_12z-GFS	Yes (hourly)	Yes	Model Run - Rolling
run KORD MR0_0z-GFS	Yes (hourly)	Yes	Horizon	Day - Rolling
KORD FH13-GFS	Yes (4 model run)	Yes	Horizon	Hour - Rolling
KORD FHH1-GFS	Yes (4 model run)	Yes	Horizon	Day - Specific
KORD FH190228-GFS	Yes (4 model run)	Yes	Horizon	Hour - Specific
KORD FH19022812-GFS	Yes (4 model run)	Yes	Horizon	Day - Rolling
run KORD FH12_12z-GFS	Yes (1 point)	Yes	Horizon	Hour - Rolling
run KORD FHH1_18z-GFS	Yes (1 point)	Yes	Valid Day	- Rolling
KORD FD1-GFS	Yes (4 model run)	Yes	Valid Day	- Rolling
run KORD FD1_12z-GFS	Yes (1 point)	Yes		

[0243] The model run and model run date retrieval method represents a ‘vanilla’ way of viewing the data. In this example, the time series may be comprised of the different model hours over that model date. The data may appear similar to a forward curve in the future. The model run and model run date retrieval may also include options such as synthetic rolling model run retrieval method and a synthetic rolling model run and model time retrieval method. The synthetic rolling model run method may programmatically include the most current model run (versus the previous or previous+1) models. The synthetic rolling model run and model time method may programmatically include the most current model run/by selected model time (versus the previous or previous+1) models.

[0244] The valid day/date-time retrieval method may allow the user to select a point in time (e.g., noon on Friday) and analyze how the forecast has changed on each model run. This retrieval method may be akin to viewing a future going towards expiry, with each model run being a session. In some examples, this retrieval method may be used for intraday (e.g., hourly) values.

[0245] The valid day or date retrieval method is similar to the date-time retrieval method, except that the dataset in the valid day/date retrieval method may include a daily aggregation and a greater number of symbols. The valid day/date retrieval method may also include an option such as a synthetic rolling valid day or date retrieval method, which may provide the ability to use indices to obtain rolling valid dates.

[0246] In some examples, the symbology may be constructed with the following elements:

Location+Model Name (e.g., GFS)+Model Run

For forecast valid days, the symbology may include:

Location+Model Name+Forecast Continuation <Roll number>

For forecast horizon days (also referred to as Target Date), the symbology may include:

Location+Model name+Forecast Horizon/Target Date <date/time>

[0247] In some examples, the symbology may support one or more types of constructions. Table 8 provides an example list of symbol types for various constructions. In some examples, it may be desirable that, similar to the continuation symbols, all metadata reflect a prompt-most dataset in a series. In Table 8, the terms horizontal, diagonal and vertical/column refers to the viewing of the symbology elements with respect to the columns and rows of the spreadsheet shown in FIG. 8B. For example, horizontal refers viewing the symbology with respect to a row, vertical refers to viewing the symbology with respect to a column and diagonal refers to viewing the symbology with respect to a diagonal line (across a combination of rows and columns).

TABLE-US-00008 TABLE 8 Example List of Symbols for Various Constructions

Symbol Type	Symbol Notes
Model Run - Rolling KORD MR0-GFS Points to core (horizontal)	Model Run - Specific KORD MR190228_12z-GFS Core (horizontal)
Model Run - Rolling run KORD MR0_0z-GFS Points to core (horizontal)	Horizon Day - Rolling KORD FH13-GFS Diagonal
Horizon Hour - Rolling KORD FHH1-GFS Diagonal	Horizon Day - Specific KORD FH190228-GFS Diagonal
Horizon Hour - Specific KORD FH19022812-GFS Diagonal	Horizon Day - Rolling run KORD FH12_12z-GFS Diagonal
Horizon Hour - Rolling run KORD FHH1_18z-GFS Diagonal	Valid Day - Rolling KORD FD1-GFS Continuations of core (vertical/column)
Valid Day - Rolling run KORD FD1_12z-GFS Continuations of core (vertical/column)	

[0248] In some examples, the symbology may include additional condition codes, including one or more of a Daily % Complete code, an Interpolated code, and a Fast/Full code. For example, “fast” and “full” selections may be handled by one or more associated condition codes. In some examples, the symbology may include additional commodity codes for different % Complete values (e.g., ¼th, ⅛th, 1/24th, etc., depending on the model and forecast day). In some examples, the “% Complete” value may be provided at 50% or higher.

[0249] In some examples, the symbology may include different selection presentation behavior depending on which condition codes are selected. In some examples, a default behavior for presenting any charts may include presenting a highest percent complete available for any point. In some examples, if a “100% complete” code is selected, then no other % complete conditions may be shown. In some examples, a default behavior for a “Fast/Full” condition code may include presenting a “latest” (typically full) (e.g., unless the full information hasn't yet arrived). In some examples, if “Full” code is selected, the “Fast” value may not be shown. In some examples, if the “Interpolated” condition code is selected as the default behavior, interpolated data may be presented. % Complete, for example (e.g., ¼ths, ⅛ths, 1/24ths, depending on the model and forecast day).

[0250] In some examples, the symbology may include condition codes for the front end (e.g., an interactive GUI, a spreadsheet, etc.), such as condition codes in “Time and Sales”, and may provide additional fields for a forecast valid hour/day. In some examples, the symbology may provide for updates to values in one or more charts. In some examples, the symbology may provide the ability to filter tick charts by “% complete” and/or interpolation by symbol. In some examples, a same symbol and different % completes may be provided in a same chart. In some examples, data may be pulled into a spreadsheet via time and/or sales information.

[0251] In some examples, the symbology may use metadata including, without being limited to, one or more of latitude (“Lat”), longitude (“Long”), altitude, elevation, a description, description

Examples of an Interactive GUI

[0257] Next, examples of an interactive GUI **900** for providing interactive tools for viewing and interacting with integrated weather and market data are described with respect to FIGS. **9A-9H**. In some examples, interactive GUI **900** may represent an example of weather impact dashboard **122** (FIG. **1**).

[0258] In some examples, interactive GUI **900** may include one or more windows, such as window **902** and window **904**. In some examples, window **902** may include user input region **906** (e.g., for providing a weather data request that may correspond to a weather symbology instruction). In some examples, window **902** may further include display region **906** for displaying weather and/or market data. In some examples, display region **906** may include an interactive display. FIGS. **9B** shows a portion of display region **910** of interactive GUI **900** illustrating an example chart. FIG. **9C** shows window **912** of interactive GUI **900** illustrating an example integrated weather and market data chart. FIG. **9D** shows window **914** of interactive GUI **900**, illustrating options for bias and bias adjusted overlays. FIG. **9E** shows window portion **916** of interactive GUI **900**, illustrating options for creating a customizable marker. FIG. **9F** shows window **918** of interactive GUI **900**, illustrating options for creating condition alerts. FIG. **9G** shows window **920** of interactive GUI **900**, illustrating options for managing custom markers. FIG. **9H** shows a portion of window **922** of interactive GUI **900**, illustrating an example grid display. It should be understood that FIGS. **9A-9H** represents a non-limiting example user interface, and that a differing interface configured in accordance with this disclosure may be provided for providing presentation and interaction with integrated weather and market data.

[0259] In some examples, interactive GUI **900** may include global controls such as time zone support and at least one weather properties section for selecting weather properties (e.g., Celsius/Fahrenheit, Mm/in (precip), M/Fect (Waves), Hpa/inhg/mmhg (pressure) and Mph/kmh/mps (wind)). In some example, interactive GUI **900** may include user-selectable options for chart presentation (e.g., an ability to add interval start/end/received to any chart (interval start=Forecast Valid), an ability to add % complete, an ability to add various conditions codes (e.g., % complete, fast/full, interpolated, not interpolated), and an ability to add various interval time values (e.g., received, start, end)).

[0260] In some examples, interactive GUI **900** may include user-selectable options for weather objects. For example, users may be able to apply various criteria to their charts through the boxes shown in user input region **906** of window **902** (FIG. **9A**). In some examples, criteria boxes can be hidden/displayed when clicking the screener button in the upper right corner (in window **902**). In some examples, if the user has entitlements for weather, and a weather symbol is entered into the top left, in some example a button will appear.

[0261] In some examples, configurable objects in a weather window may include location region control (this is a type-ahead search only on weather symbols), model run (e.g., 0z, 6z, 12z, 18z, current model run, current+6 hrs ago, current+12 hrs ago, current+18 hrs ago, current+24 hrs ago), model (e.g., GFS, GEFS, etc.), symbol granularity (daily, synthetic hourly, raw data), climatology overlay (e.g., NWS 30 yr, Trail 30 yr, Trail 15 yr, Highest, 1+std, Mean, -1 std, Lowest), and studies (e.g., accumulate, show with bias, show only bias, summary stats only).

[0262] In some examples, interactive GUI **900** may provide an ability to save a criteria and view modify existing weather window views. In some examples, interactive GUI **900** may be configured to provide a display of data based on criteria that can be filtered through user entry, selection, suggested values, other, to filter results based on users' needs. Example controls available may include checkboxes, radio buttons, ranges and/or manually entered alpha/numerical values. In some examples, users may be permitted to temporarily turn on/off criteria by clicking on, for example, a green light at upper left of criteria box for what if scenarios. In some examples, criteria boxes may be collapsed to offer more screen real estate for a data view. This operation may be performed, for example, by clicking a double arrow on the bottom left of toolbar. Criteria filtered entries may still

visible in collapsed mode. In some examples, users may be permitted to still modify criteria filters in collapsed mode by clicking on highlighted text. In some examples, users may save criteria sets for easy recall of filtered sets.

[0263] In some examples, interactive GUI **900** may provide options for studying the impact of weather data on market data. For example, interactive GUI **900** may provide climatology overlays. In some examples, statistical ranges for the overlays may take a portion of or all weather series in a chart as input. Examples of the statistical ranges may include a highest high, an upper 1 sigma, a lower 1 sigma, a lowest low and a mean. In some examples, interactive GUI **900** may also provide options for bias and bias adjusted overlays (see, FIG. **9D**).

[0264] In some examples, interactive GUI **900** may provide user-selectable options for markers enhancement. In some examples, the markers enhancement options may include the ability to create a new customizable marker (e.g., when a user clicks “Customize” from the Markers drop down, they may see a window similar to custom alerts). (See FIGS. **9E** and **9F**.) In some example, the markets enhancement options may include a way to toggle between all events where the expression is true, and when the first event occurs (cross over). For example, If KORD DIFF-GFS>4 AND KORD Change-GFS>4. In some examples, markers may use an interval start for cross reference. In some examples, markers may contain metadata on hover (e.g., value, expression, time received). In some examples, a window may be provided for managing custom markers (see FIG. **9G**).

[0265] In some examples, interactive GUI **900** may be configured to provide a user with options to generate a watchlist. One example use case is using the FV day type symbols to pull in GWDD as the data arrives. In some examples, interactive GUI **900** may include additional weather colorization logic for the watchlists.

[0266] In some examples, interactive GUI **900** may be configured to generate one or more grids. In some examples, a grid may be the primary location for creating displays of this weather data. In some examples, the grid may include an ability to pull out history for any hourly point, daily point, and any historical field. In some examples, additional grid syntax may be provided to filter on new weather condition codes (e.g., interpolated, % complete, fast vs. full). In some examples, additional grid syntax may be provided to request data by different timeline indices (e.g., start, end, received). In some examples, additional grid coloration logic may be provided for weather (e.g., which may be user defined in weather global parameters). See FIG. **9H** for an example of a grid.

[0267] Referring next to FIGS. **10A-10K**, examples of an interactive GUI **1000** for providing interactive tools for viewing and interacting with integrated weather and market data are described, according to another aspect of the present disclosure. In some examples, interactive GUI **1000** may represent an example of weather impact dashboard **122** (FIG. **1**). It should be understood that FIGS. **10A-10K** represent a non-limiting example user interface, and that a differing interface configured in accordance with this disclosure may be provided for providing presentation and interaction with integrated weather and market data.

[0268] In some examples, interactive GUI **1000** may include weather landing page **1002**, as shown in FIG. **10A**. Weather landing page **1002** may include various regions **1004-1014** (shown in detail in respective FIGS. **10B-10F**) for viewing and interacting with user-customizable weather and market data in real-time (including weather data integrated with market data), and for creating user-customizable watchlists and alerts. In some examples, weather landing page **1002** may include weather alpha region **1004** (shown in FIG. **10B**), map region **1006** (shown in FIG. **10C**), model status region **1008** (shown in FIG. **10D**), watchlist region **1010** (shown in FIG. **10F**), drill around region **1012** (shown in FIG. **10F**) and alerts region **1014** (shown in FIG. **10E**). Weather landing page **1002** may also include one or more additional selectable tabs (e.g., hourly meteorology, hourly guidance, etc.) as shown in FIG. **10A**) which may open one or more additional windows, pop-up windows and/or regions in weather landing page **1002**.

[0269] Referring to FIG. **10B**, in some examples, weather alpha region **1004** may be configured to

present live streaming weather-alpha data for display and user-interaction, including integrated live streaming weather data and live streaming market price data. Weather alpha region **1004** may also be configured to receive user input (e.g., via one or more user buttons, drop down menus, text boxes and the like). In some examples, hovering over a location by a pointing device (e.g., a mouse) may cause weather alpha region **1004** to present additional information. In some examples, weather alpha region **1004** may be configured to present live streaming weather and price in accordance with a specific location (e.g., a specific trading location) in an graphical format. In some examples, the weather data may be expressed in one or more “trader” time periods (e.g., NG weeks, Calendar weeks, etc.) to better match any time periods for transacting in any desired financial instruments.

[0270] In general, weather alpha region **1004** may be configured to depict how weather forecasts (e.g., changing weather perceptions and/or anomalous weather) may move commodity prices, in real-time. Weather alpha region **1004**, may be used, together with the weather symbology of the present disclosure to create unique (and user-customizable) financial weather workflows and provide unique transformational integrated weather and price visualizations. Such unique and customizable integrated weather and price visualizations are exclusive to the present disclosure and do not exist in conventional systems. Although it is generally known that weather versus price relationships may exist, challenges exist in quantifying market efficiencies/inefficiencies of market price digesting this information, as well as any frequency of occurrences, a market breadth of weather impacts and a magnitude of the price changes related to changing weather perceptions. Interactive GUI **1000** (including weather alpha region **1004**), through the unique integrated and customized presentation of live weather and market data, significantly improves this type of weather/price analysis and provides such data and information in an easily accessible manner for various types of users (e.g., in the financial community), and brings a once analog weather market into a streamlined digital workflow.

[0271] In some examples, weather alpha region **1004** may include options for providing weather alpha information according to an alpha micro presentation or an alpha macro presentation. For an alpha micro presentation, weather alpha region **1004** may provide the fastest and most detailed views of changing weather perspectives (e.g. from key government entity forecasts) overlaid with real-time market prices. In some examples, an alpha micro presentation may use preliminary daily values processed in real-time from hourly forecast data, to deliver an early view of forecasted daily temps & GWDD, minutes in advance of alternate weather providers. For an alpha macro presentation, weather alpha region **1004** may provide a comprehensive qualitative review over an entire trading day (or multiple trading days) showing how various weather forecast models (including various model runs) have influenced market prices (for any instrument). In some examples, the alpha macro presentation may provide a quick qualitative view of a directional relationship between weather and price data.

[0272] Referring to FIG. **10C**, map region **1006** may be configured to present one or more weather maps, including live streaming weather maps. In some examples, the weather map(s) may be presented in a graphical format and may include various colorizations (e.g., to indicate weather information such as temperature gradations). In some examples, map region **1006** may provide user input options for selecting and customizing various types of weather maps. The presented weather maps in map region **1006**, working in concert with the presented weather and price information in weather alpha region **1004**, may allow the user to view big picture impacts of weather, for example, the impacts of weather about to enter into a region of interest of a weather map. In some examples, map region **1006** may deliver weather map(s) within seconds after government entity weather forecasts are made available into the public domain. In some examples, the weather maps may include maps related to one or more of hourly weather data, daily weather data, multi-day weather data, seasonal/performance data, weather anomaly data and change of weather. The weather maps may include one or more comprehensive maps covering the Earth's sub-surface, surface, upper air,

along with liquid/frozen precipitation, renewables, etc.

[0273] In some examples, weather map(s) presented in map region **1006** may depend on a particular weather workflow. In a guidance workflow, weather map(s) may be presented in such a manner to easily compare a number of weather forecasts for relative agreement/disagreement (e.g., a forecast risk). In a progression workflow, weather map(s) may be presented in such a manner to hold target data constant to view how all weather forecasts have changed over time for a specific target date. In a performance workflow, weather map(s) may be presented in a manner to help understand weather forecast accuracy (e.g., by exploring warmer/colder bias or mean absolute error).

[0274] In some examples, seasonal map(s) may be configured in a manner to easily compare and contrast longer range weather forecasts (e.g., covering a balance-of-the-month, a prompt month, a next two seasons, etc.). In some examples, map region **1006** may include user options for selecting various streaming weather maps (e.g., geographic region, time period (e.g., hourly, daily, multi-day), model run date, weather map type and/or popular maps). In some examples, options for a geographic region may include options for global, regional, tropics and GIS.

[0275] Referring to FIG. **10D**, in some examples, model status region **1008** may provide real-time (e.g., up-to-date) status information of various weather model downloads. In some examples, model status region **1008** may include user-customizable options for presentation of model status information. In some examples, model hours may be organized into Forecast Valid Days (rows) based upon a selected time zone, and may display percentage (%) complete for the particular model. In general, the model status information presented in model status region **1008** may provide a snapshot of the weather models, to indicate when the weather forecast models have loaded and whether there are any delays and/or missing data. In some examples, the status information may be displayed in various colorizations. For example, for weather models/model runs, a green line may indicate the model is “currently running”. A green status value may indicate that a specific model hour has arrived “on-time” as expected (e.g., based upon timestamps from the trailing fourteen days). A yellow status value may indicate that a specific model hour arrived slower than a trailing (for example) 14 day average (e.g., outside 1 standard deviation). A red status value may indicate that the specific model hour should have arrived (e.g. at least five minutes ago) and is still missing.

[0276] Referring to FIG. **10F**, in some examples, watchlist region **1010** may provide one or more user-input options for creating a user-customizable watchlist of one or more selectable financial instruments that may be linked to a desired (e.g., optimal) weather location. In some examples, drill around region **1012** may include one or more drill-Around boxes associated with one or more financial instruments. The drill-around box(es) may provide one or more options related to one or more desired weather symbologies associated with the particular financial instrument(s).

[0277] Referring to FIG. **10E**, in some examples, alerts region **1014** may include user-input options for creating customizable weather alerts, such as changing weather forecast(s) that may be considered by the particular user to be “significant.” In some examples, alerts region **1014** may present active weather alerts, options to allow a user to add new weather alerts and/or options to view a history of one or more (including all) weather alerts which have been triggered. The information provided in alerts region **1014** may allow users to more confidently cover more weather influenced instruments with greater precision and confidence. In some examples, an alert may direct a user to specifically designed workbook when the alert is triggered.

[0278] Although exemplary regions are depicted in FIG. **10A**, alternate configurations for the regions are envisioned. For example, a graphical user interface may contain more or fewer regions. Additionally, the regions may be reorganized in any manner and display other pertinent information.

[0279] FIG. **10G** illustrates an example weather alert set up window **1016** (window **1016** herein) of interactive GUI **1000**. Window **1016** represents a wizard for creating user-customizable weather alerts. In general, customizable weather alerts may allow a user to monitor streaming weather

forecasts, and identify (user-specified) changes expected (by the user) to have a market impact. Window **1016** may include one or more user input regions for adding a new (user-customizable) weather alert (which may be presented in alerts region **1014** (FIG. **10E**)). In a non-limiting example, window **1016** may include location input region **1018**, alert threshold input region **1020**, weather model selection region **1022** and notification selection region **1024**.

[0280] Location input region **1018** may provide a user input region for user input of one or more desired weather location. Alert threshold input region **1020** may include options for user input of one or more thresholds, to create one or more conditions for triggering an alert. In some examples, alert threshold input region **1020** may include options for scanning through historical data to select appropriate alert thresholds and/or a notes region for user input for indicating special notes (e.g., bullish/bearish) linked to an financial instrument. In some examples, weather model selection region **1022** may include user input options for selecting one or more models, one or more model runs, one or more forecast periods, forecast valid days to scan and at least one percentage complete value to utilize for the forecast. In some examples, notification selection region **1024** may include user input options for selecting one or more days a user desires the alert(s) to be active.

[0281] FIG. **10H** illustrates an example weather perspective selection window **1026** of interactive GUI **1000** when workflow selector region **1028** is selected for a guidance workflow (also referred to herein as guidance window **1026**). In general, weather perspective selection window **1026** represents a weather wizard for creating user-customizable charts for various weather perspectives (e.g., weather workflow(s), weather alpha(s)). The Guidance workflow may be configured to present weather data in a manner to help the user to understand forecast risk (e.g., relative agreement & disagreement among forecasts). The guidance workflow may provide another unique perspective to view same day weather forecasts, and how current weather forecasts relate to previous years.

[0282] Guidance window **1026** may include one or more user input regions for creating a user-customized guidance workflow. In a non-limiting example, guidance window **1026** may include location input region **1030** for selecting one or more weather locations, time interval input region **1032** for selecting hourly or daily time intervals, weather model selection region **1034** and observations selection region **1036**. In some examples, weather model selection region **1034** may include user input for selecting one or more weather forecast models, one or more model runs, any ensemble statistics and one or more climatology options. Observations selection region **1036** may include user input options for observing the data, including real-time observations and historical weather forecasts.

[0283] In some examples, workflow selector region **1028** may include a weather workflow option for seasonal analysis (not shown). The seasonal analysis workflow may be configured to present data to easily compare years whose weather pattern(s) maybe similar to a current season, along with various climatologies. In some examples, the seasonal analysis workflow may be configured to overlay long range weather forecasts to gain a unique future perspective, which forecasts may be monitored in real-time as the forecast models are updated.

[0284] FIG. **10I** illustrates an example weather perspective selection window **1038** of interactive GUI **1000** when workflow selector region **1028** is selected for an accumulation workflow (referred to herein as accumulation window **1038**). In general, accumulation window **1038** may include user input regions similar to guidance window **1026**, but with different options associated with creating an accumulation workflow. In general, the accumulation workflow may be configured to present data to provide a seasonal perspective to accumulated snow, precipitation, degree days, etc., to help analyze general market trends (e.g., seasons being above/below average).

[0285] FIG. **10J** illustrates an example weather perspective selection window **1040** of interactive GUI **1000** when workflow selector region **1028** is selected for a progression workflow (referred to herein as progression window **1040**). In general, progression window **1040** may include user input regions similar to guidance window **1026**, but with different options associated with creating a

progression workflow. In general, a progression workflow may be configured to present data to help users understand “convergence” of weather forecasts (e.g. when forecasts begin to agree with each other) as well as run-to-run volatility and mean reversion information.

[0286] FIG. **10K** illustrates an example weather perspective selection window **1042** of interactive GUI **1000** when workflow selector region **1028** is selected for an alpha micro workflow (referred to herein as alpha micro window **1040**). In general, alpha micro window **1042** may include user input regions similar to guidance window **1026**, but with different options associated with creating an alpha micro workflow. In some examples, workflow selector region **1028** may also include an alpha macro window for creating a user-customized alpha macro workflow (not shown). In some examples, the alpha macro workflow may provide a quick qualitative view of the directional relationship between weather and price data over multiple model runs and/or multiple days. In general, a weather alpha workflow (e.g., alpha macro, alpha micro) may be configured to present forecasted GWDD from each weather forecast model, as it is released in the public domain vs price data. In some examples, the weather alpha workflow may be configured for such that a user may quickly scan across many different weather models and model runs and identify if the weather may be driving market price(s).

Computer Architecture

[0287] Systems and methods of the present disclosure may include and/or may be implemented by one or more specialized computers including specialized hardware and/or software components. For purposes of this disclosure, a specialized computer may be a programmable machine capable of performing arithmetic and/or logical operations and specially programmed to perform the functions described herein. In some embodiments, computers may comprise processors, memories, data storage devices, and/or other commonly known or novel components. These components may be connected physically or through network or wireless links. Computers may also comprise software which may direct the operations of the aforementioned components. Computers may be referred to as servers, personal computers (PCs), mobile devices, and other terms for computing/communication devices. For purposes of this disclosure, those terms used herein are interchangeable, and any special purpose computer particularly configured for performing the described functions may be used.

[0288] Computers may be linked to one another via one or more networks. A network may be any plurality of completely or partially interconnected computers wherein some or all of the computers are able to communicate with one another. It will be understood by those of ordinary skill that connections between computers may be wired in some cases (e.g., via wired TCP connection or other wired connection) or may be wireless (e.g., via a WiFi network connection). Any connection through which at least two computers may exchange data can be the basis of a network.

Furthermore, separate networks may be able to be interconnected such that one or more computers within one network may communicate with one or more computers in another network. In such a case, the plurality of separate networks may optionally be considered to be a single network.

[0289] The term “computer” shall refer to any electronic device or devices, including those having capabilities to be utilized in connection with an electronic information/transaction system, such as any device capable of receiving, transmitting, processing and/or using data and information. The computer may comprise a server, a processor, a microprocessor, a personal computer, such as a laptop, palm PC, desktop or workstation, a network server, a mainframe, an electronic wired or wireless device, such as for example, a telephone, a cellular telephone, a personal digital assistant, a smartphone, an interactive television, such as for example, a television adapted to be connected to the Internet or an electronic device adapted for use with a television, an electronic pager or any other computing and/or communication device.

[0290] The term “network” shall refer to any type of network or networks, including those capable of being utilized in connection with the systems and methods described herein, such as, for example, any public and/or private networks, including, for instance, the Internet, an intranet, or an

extranet, any wired or wireless networks or combinations thereof.

[0291] The term “computer-readable storage medium” should be taken to include a single medium or multiple media that store one or more sets of instructions. The term “computer-readable storage medium” shall also be taken to include any medium that is capable of storing or encoding a set of instructions for execution by the machine and that causes the machine to perform any one or more of the methodologies of the present disclosure.

[0292] FIG. **11** illustrates a functional block diagram of a machine in the example form of computer system **1100** within which a set of instructions for causing the machine to perform any one or more of the methodologies, processes or functions discussed herein may be executed. In some examples, the machine may be connected (e.g., networked) to other machines as described above. The machine may operate in the capacity of a server or a client machine in a client-server network environment, or as a peer machine in a peer-to-peer (or distributed) network environment.

[0293] The machine may be any special-purpose machine capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that machine for performing the functions describe herein. Further, while only a single machine is illustrated, the term “machine” shall also be taken to include any collection of machines that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein. In some examples, one or more of components **112-128** of DID system **102**, weather data source(s) **104**, market data source(s) **106** and/or user device(s) **108** may be implemented by a specialized machine, particularly programmed to perform certain functions, such as the example machine shown in FIG. **11** (or a combination of two or more of such machines).

[0294] Example computer system **1100** may include processing device **1102**, memory **1106**, data storage device **1110** and communication interface **1112**, which may communicate with each other via data and control bus **1118**. In some examples, computer system **1100** may also include display device **1114** and/or user interface **1116**.

[0295] Processing device **1102** may include, without being limited to, a microprocessor, a central processing unit, an application-specific integrated circuit (ASIC), a field programmable gate array (FPGA), a digital signal processor (DSP) and/or a network processor. Processing device **1102** may be configured to execute processing logic **1104** for performing the operations described herein. Processing device **1102** may include a special-purpose processing device specially programmed with processing logic **1104** to perform the operations described herein.

[0296] Memory **1106** may include, for example, without being limited to, at least one of a read-only memory (ROM), a random access memory (RAM), a flash memory, a dynamic RAM (DRAM) and a static RAM (SRAM), storing computer-readable instructions **1108** executable by processing device **1102**. Memory **1106** may include a non-transitory computer readable storage medium storing computer-readable instructions **1108** executable by processing device **1102** for performing the operations described herein. For example, computer-readable instructions **1108** may include operations performed by components **112-128** of DID system **102**, including operations shown in FIGS. **3A** and **3B**. Although one memory device **1106** is illustrated in FIG. **11**, in some examples, computer system **1100** may include two or more memory devices (e.g., dynamic memory and static memory).

[0297] Computer system **1100** may include communication interface device **1112**, for direct communication with other computers (including wired and/or wireless communication) and/or for communication with a network. In some examples, computer system **1100** may include display device **1114** (e.g., a liquid crystal display (LCD), a touch sensitive display, etc.). In some examples, computer system **1100** may include user interface **1116** (e.g., an alphanumeric input device, a cursor control device, etc.).

[0298] In some examples, computer system **1100** may include data storage device **1110** storing instructions (e.g., software) for performing any one or more of the functions described herein. Data storage device **1110** may include a non-transitory computer-readable storage medium, including,

without being limited to, solid-state memories, optical media and magnetic media.

[0299] While the present disclosure has been discussed in terms of certain embodiments, it should be appreciated that the present disclosure is not so limited. The embodiments are explained herein by way of example, and there are numerous modifications, variations and other embodiments that may be employed that would still be within the scope of the present disclosure.

Claims

1. A method comprising: providing a data distribution system in communication with one or more weather data source systems, one or more market data source systems and at least one user device via at least one network, the data distribution system comprising a processing device and a memory storing computer-readable instructions; collecting, by the data distribution system, weather data from among the one or more weather data source systems, the weather data comprising a plurality of formats; reformatting, by the data distribution system, the weather data to generate a weather forecast dataset having a singular format, wherein the reformatting comprises segmenting the weather data into a plurality of weather data segments to form a time series dataset configured to be processed by a time series server, each weather data segment of the plurality of weather data segments comprising a respective exchange time; collecting, by the data distribution system, financial market data from among the one or more market data source systems, the financial market data associated with one or more exchange times and configured to be processed by the time series server; integrating, by the time series server, the weather forecast dataset and a portion of the collected financial market data in a manner indicating a correlation between the weather forecast dataset and the portion of the collected financial market data by matching each respective exchange time of the plurality of weather data segments and the one or more exchange times of the portion of the collected financial market data; generating, by the data distribution system, an interactive graphical user interface (GUI) for display on the at least one user device; displaying, via the interactive GUI, a presentation package that includes the weather forecast dataset integrated with said portion of the collected financial market data, the presentation package indicating the correlation between the weather forecast dataset and the portion of the collected financial market data; monitoring, by the data distribution system, changes to one or more of the weather data and the financial market data; updating the presentation package on the interactive GUI concurrent with one or more of user input received via the at least one user device and any of the monitored changes to one or more of the weather data and the financial market data; determining, by the data distribution system, that the monitored changes to one or more of the weather data and the financial market data satisfy one or more user-customized thresholds received by the interactive GUI via the at least one user device; generating, by the data distribution system, at least one alert indicating satisfaction of the one or more user-customized thresholds; and displaying, via the interactive GUI, the at least one alert.

2. The method of claim 1, further comprising: storing, in at least one database accessible by the data distribution system, a weather symbology comprising one or more symbol elements linked to segments of the collected weather data and one or more pre-defined rules for generating weather symbology instructions having a pre-defined instruction structure.

3. The method of claim 2, further comprising: receiving, via the interactive GUI, input comprising a weather data request from the at least one user device, and determining, by the data distribution system, a weather symbology instruction based on at least one requested symbol element indicated in the received weather data request and in accordance with the one or more pre-defined rules.

4. The method of claim 3, wherein the weather forecast dataset is generated based, at least in part, on the weather symbology instruction.

5. The method of claim 2, wherein the one or more symbol elements are associated with one or more of at least one weather forecast model, at least one weather observation location and at least

one predetermined weather workflow structure.

6. The method of claim 5, wherein the at least one predetermined weather workflow structure includes one or more of at least one weather alpha, a guidance perspective, a progression perspective and an accuracy perspective.

7. The method of claim 5, wherein the interactive GUI includes at least one wizard window for user-customization of the at least one predetermined weather workflow structure.

8. The method of claim 3, wherein the weather symbology further includes one or more weather variables comprising at least one of predetermined fields and predetermined conditions, the weather symbology instruction further determined based on at least one requested weather variable among the one or more weather variables received in the weather data request.

9. The method of claim 3, wherein the weather symbology instruction is included in the weather data request.

10. The method of claim 3, wherein the weather data request is converted to the weather symbology instruction.

11. The method of claim 3, further comprising: extracting, by the data distribution system, at least one segment among the segments of the collected weather data corresponding to the at least one requested symbol element in the weather symbology instruction.

12. The method of claim 3, wherein the weather forecast dataset includes at least one weather forecast stitched together from among the collected weather data in accordance with the weather symbology instruction.

13. The method of claim 1, wherein the collected weather data includes weather observation data and weather forecast model data.

14. The method of claim 13, wherein the weather observation data includes at least one of Meteorological Aerodrome Report (METAR) data and Climate Forecast System Reanalysis (CFS-R) data.

15. The method of claim 13, wherein the weather forecast model data is associated with at least one of a Global Forecast System (GFS), a GFS ensemble (GFS ENS), a GFS ensemble Extension (GFS ENS EXT), a Climate Forecast System (CFS) ensemble, a European Center for Medium-Range Weather Forecast (ECMWF), a ECMWF ensemble (ECMWF ENS), a ECMWF ensemble Extension (EXCMWF ENS EXT), a ECMWF Seasonal (SEAS) and a ECMWF Seasonal Extension (SEAS EXT).

16. The method of claim 1, wherein the collected weather data and the collected financial market data includes at least one of streaming data and static data.

17. The method of claim 1, further comprising: displaying, via the interactive GUI, at least one weather map generated from among the collected weather data.

18. The method of claim 1, further comprising: exposing, by the data distribution system, the presentation package to at least one application programming interface (API).

19. The method of claim 1, wherein the interactive GUI is configured to operate on at least one of a trading desktop, a mobile application and a spreadsheet application.

20. The method of claim 1, wherein the weather data comprises one or more weather data streams received via at least one data feed, and wherein the financial market data comprises one or more financial market data streams received via the at least one data feed, the method further comprising: segmenting, by the data distribution system, the received one or more weather data streams based on one or more pre-defined symbol elements to form at least one segmented weather data stream; extracting, by the data distribution system, one or more segments of the at least one segmented weather data stream to form at least one extracted weather stream; generating, by the data distribution system, a real-time weather dataset from the at least one extracted weather stream; integrating, by the data distribution system, the real-time weather dataset with the received one or more financial market data streams to form part of an integrated presentation package; at least one of: exposing, by the data distribution system, the integrated presentation package to at least one

application programming interface (API) and displaying the integrated presentation package via the interactive GUI on a display of the at least one user device; and updating the integrated presentation package concurrent with changes to one or more of the one or more weather data streams and the one or more financial market data streams, wherein the integrated presentation package further includes an interactive presentation of one or more of at least one weather forecast map, at least one weather alpha region indicating a combination of weather forecast data and financial market data, at least one weather perspective chart associated with a predetermined weather perspective workflow, at least one model status region, at least one watchlist region, at least one alerts region, at least one chart, at least one table, at least one weather grid, at least one analytic value and at least one user input window for customizing the interactive presentation.

21. The method of claim 20, further comprising: creating, by the data distribution system, the real-time weather dataset in accordance with a weather symbology instruction.

22. The method of claim 21, wherein the weather symbology instruction further includes one or more weather variables comprising at least one of predetermined fields and predetermined conditions, the weather symbology instruction being created in accordance with at least one weather variable among the one or more weather variables.

23. The method of claim 20, wherein the integrated presentation package is configured to be launched in at least one of a trading desktop, a mobile application and a spreadsheet application.

24. The method of claim 1, wherein the presentation package further includes an interactive presentation of one or more of at least one weather forecast map, at least one weather alpha region indicating a combination of weather forecast data and the collected financial market data, at least one weather perspective chart associated with a predetermined weather perspective workflow, at least one model status region, at least one watchlist region, at least one alerts region, at least one chart, at least one table, at least one weather grid, at least one analytic value and at least one user input window for customizing the interactive presentation.

25. The method of claim 1, wherein the presentation package comprises at least one alerts region for creating one or more customizable alerts including the at least one alert.
